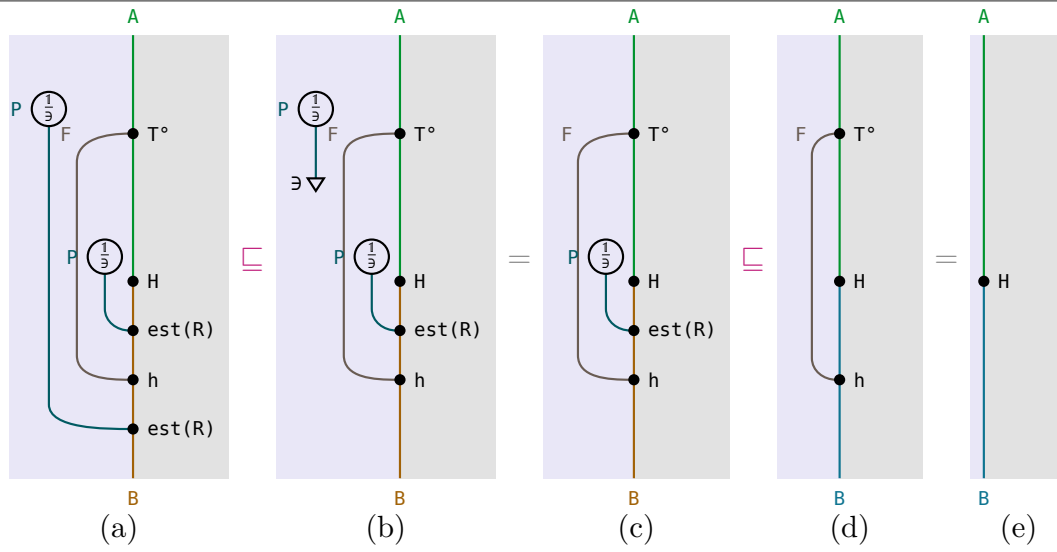


# Companion to Bird & de Moor, Algebra of Programming



$$\frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R) \sqsubseteq H \quad (9.2)$$

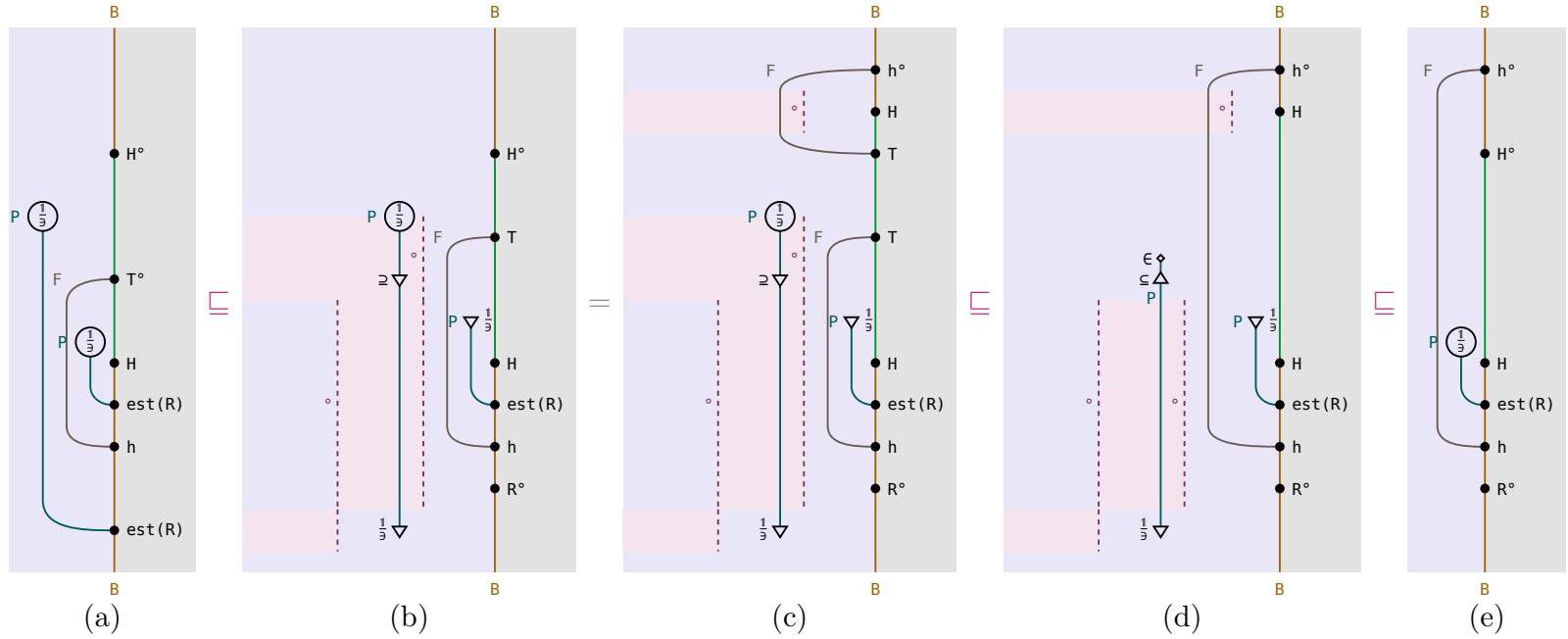
taking the input apart every way  $T$  allows ( $\frac{T^\circ}{\exists}$ ), solving each part by  $M$  and keeping an optimum ( $P(F(M)h) \text{est}(R)$ )  
returns only what  $H$  returns



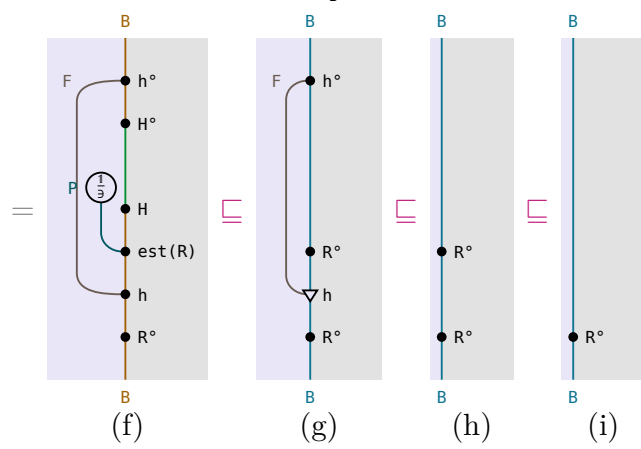
|     |                                                                                    |                                                 |
|-----|------------------------------------------------------------------------------------|-------------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R)$    | $P(S) \text{est}(R) \sqsubseteq \exists S$      |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{est}(R)) h$ | $\frac{R}{\exists} \exists = R$                 |
| (c) | $= T^\circ F(\frac{H}{\exists} \text{est}(R)) h$                                   | $\frac{R}{\exists} \text{est}(R) \sqsubseteq S$ |
| (d) | $\sqsubseteq T^\circ F(H) h$                                                       | $T^\circ F(H) h = H$                            |
| (e) | $= H$                                                                              |                                                 |

$$\text{Map}(h) \wedge F(R^\circ) \ h \sqsubseteq h \ R^\circ \wedge R^\circ \ R^\circ \sqsubseteq R^\circ \implies H^\circ \xrightarrow{\frac{T^\circ}{3}} P(F(\frac{H}{3} \text{ est}(R)) \ h) \text{ est}(R) \sqsubseteq R^\circ \quad (9.3)$$

for every  $\mathbf{b}$  that  $\mathbf{H}$  returns from an input, the step  $\frac{T^\circ}{3} P(F(\mathbf{M})h) \text{ est}(R)$  returns from that input only  $\mathbf{b}'$  with  $\mathbf{R}$  relating  $\mathbf{b}'$  to  $\mathbf{b}$



|     |                                                                                                                             |                                                                    |
|-----|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| (a) | $H^\circ \xrightarrow{\frac{T^\circ}{3}} P(F(\frac{H}{3} \text{ est}(R)) \ h) \text{ est}(R)$                               | $P(S) \text{ est}(R) \sqsubseteq \epsilon \setminus (S \ R^\circ)$ |
| (b) | $\sqsubseteq H^\circ \xrightarrow{\frac{T^\circ}{3}} (\epsilon \setminus (F(\frac{H}{3} \text{ est}(R)) \ h \ R^\circ))$    | $H^\circ = h^\circ F(H^\circ) T$                                   |
| (c) | $= h^\circ F(H^\circ) T \xrightarrow{\frac{T^\circ}{3}} (\epsilon \setminus (F(\frac{H}{3} \text{ est}(R)) \ h \ R^\circ))$ | $T \xrightarrow{\frac{T^\circ}{3}} \epsilon$                       |
| (d) | $\sqsubseteq h^\circ F(H^\circ) \epsilon \in (\epsilon \setminus (F(\frac{H}{3} \text{ est}(R)) \ h \ R^\circ))$            | $S \ (S \setminus R) \sqsubseteq R$                                |
| (e) | $\sqsubseteq h^\circ F(H^\circ) F(\frac{H}{3} \text{ est}(R)) \ h \ R^\circ$                                                |                                                                    |



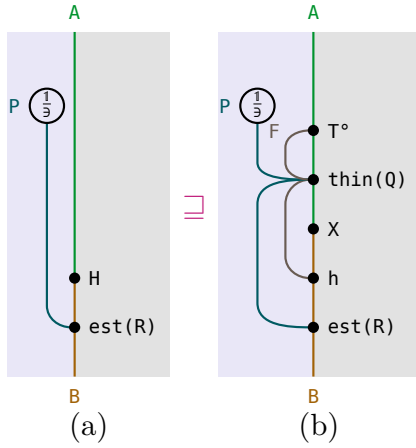
|     |                                                                               |                                                                           |
|-----|-------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| (f) | $= h^\circ F(H^\circ \xrightarrow{\frac{H}{3}} \text{ est}(R)) \ h \ R^\circ$ | $S^\circ \xrightarrow{\frac{S}{3}} \text{ est}(R) \sqsubseteq R^\circ$    |
| (g) | $\sqsubseteq h^\circ F(R^\circ) \ h \ R^\circ$                                | $\text{Map}(f)$                                                           |
| (h) | $\sqsubseteq R^\circ \ R^\circ$                                               | $\implies F(R) \ f \sqsubseteq f \ R \iff f^\circ F(R) \ f \sqsubseteq R$ |
| (i) | $\sqsubseteq R^\circ$                                                         | $R^\circ \ R^\circ \sqsubseteq R^\circ$                                   |

**Map(h)  $\wedge$  F(R<sup>o</sup>)  $h \sqsubseteq h R^o \wedge R^o R^o \sqsubseteq R^o \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^o}{3} \text{thin}(Q)$**  Theorem 9.2

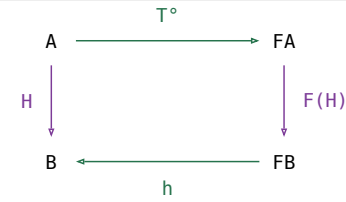
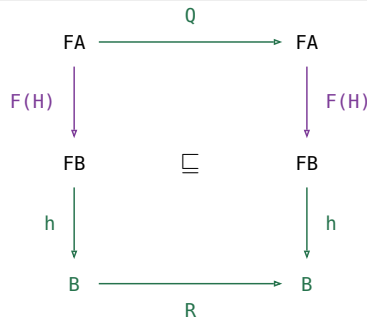
$$P(F(X) h) \text{est}(R)) \sqsubseteq \frac{H}{3} \text{est}(R)$$

an optimum over everything **H** returns is reached by taking the input apart every way **T** allows, dropping the parts that can never win, solving each of the rest and keeping one optimum

$$H \triangleq (T)^o \circ (h) \quad H \triangleq (T)^o \circ (h)$$



|     |                                                                    |                                                                                                                                                                                                                                                                                                 |
|-----|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{H}{3} \text{est}(R)$                                        | $QF(H)h \sqsubseteq F(H)hR, h \text{ monotonic on } R \text{ — } Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R R \sqsubseteq R$<br>$\Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R), X \sqsubseteq \text{est}(R) \Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^o$ |
| (b) | $\sqsubseteq \frac{T^o}{3} \text{thin}(Q) P(F(X) h) \text{est}(R)$ |                                                                                                                                                                                                                                                                                                 |



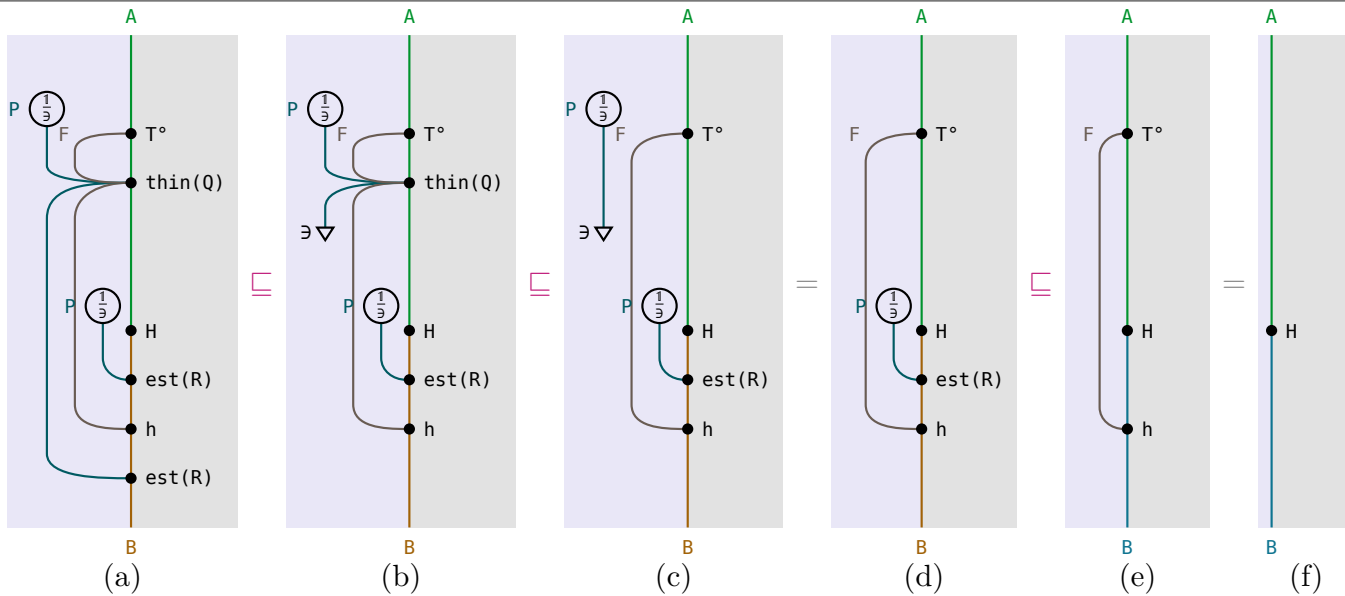
$T^o F(H)h = H$ : taking the input apart once by  $T^o$ , solving the parts by  $F(H)$  and assembling by  $h$  returns exactly what  $H$  returns

$QF(H)h \sqsubseteq F(H)hR$ : every result of thinning by  $Q$ , solving the parts by  $F(H)$  and assembling by  $h$  is reached by  $R$  from a result of solving by  $F(H)$  and assembling by  $h$  without  $Q$

$$\frac{T^o}{3} \text{thin}(Q) P(F(\frac{H}{3} \text{est}(R)) h) \text{est}(R) \sqsubseteq H$$

(9.1.1e)

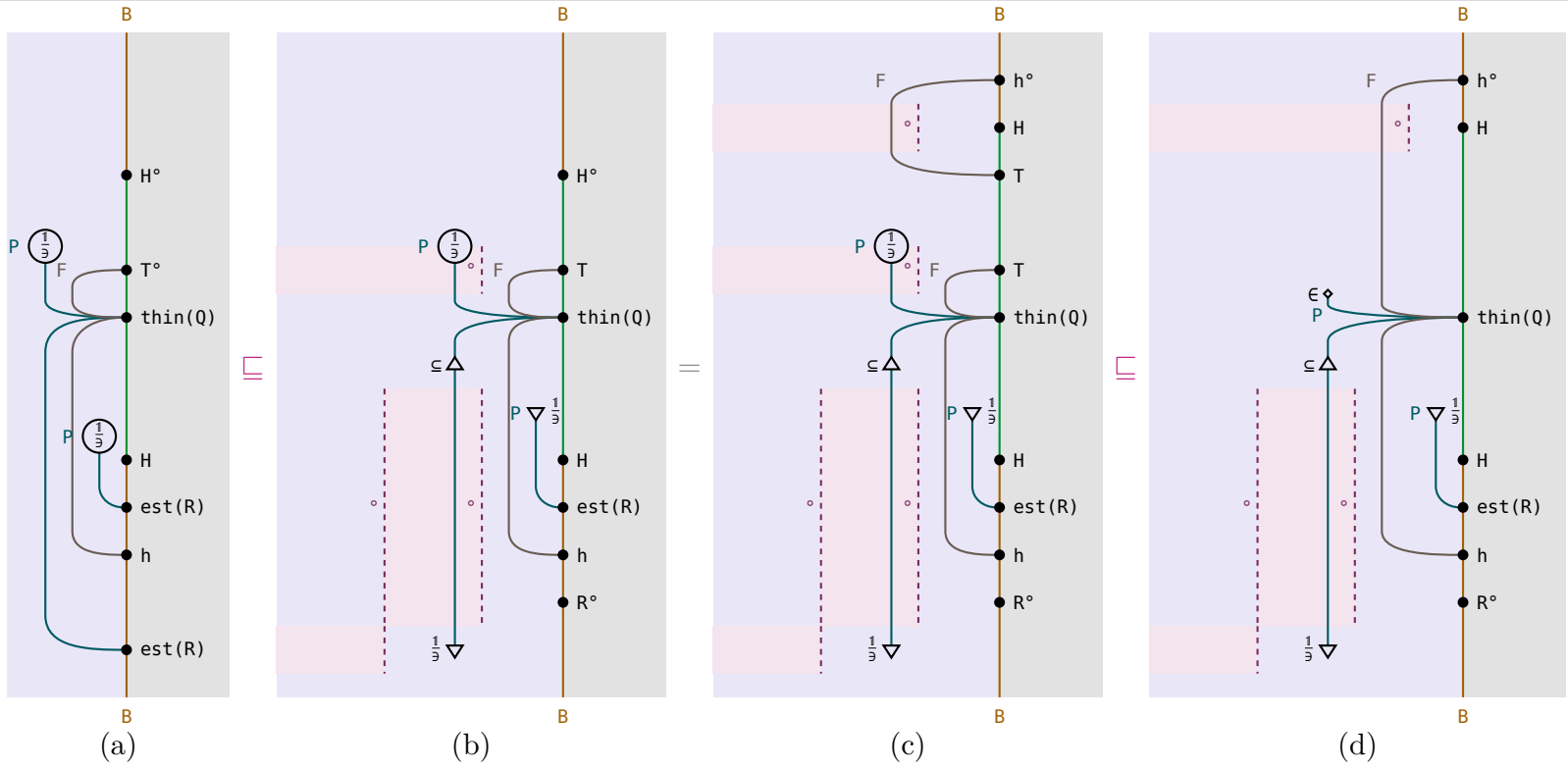
taking the input apart every way **T** allows ( $\frac{T^o}{3}$ ), dropping the parts **thin(Q)** rejects, solving each rest by **M** and keeping an optimum returns only what **H** returns



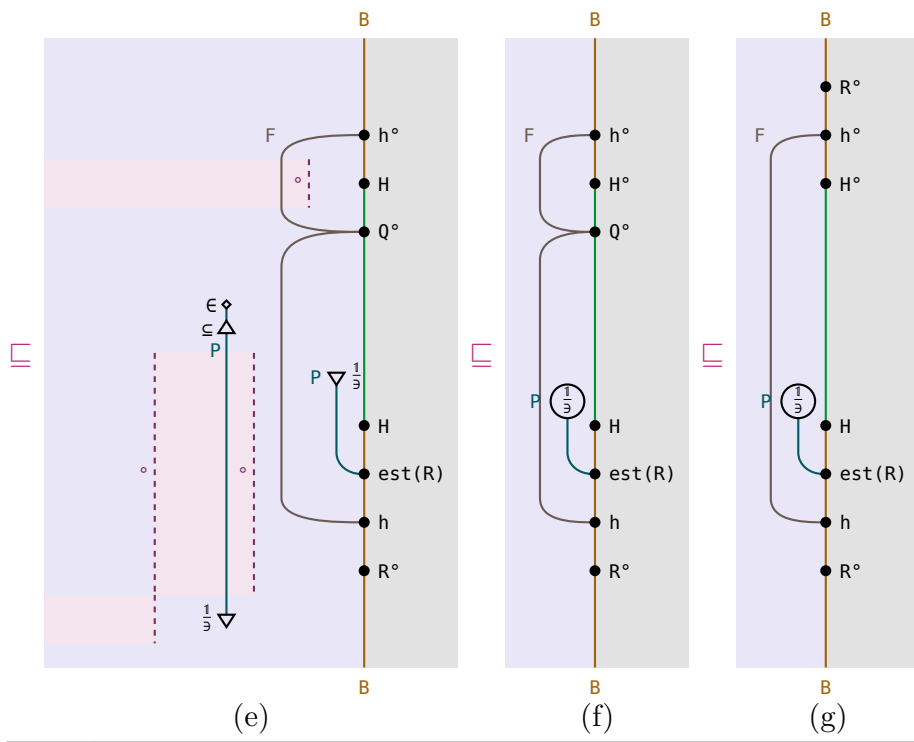
|     |                                                                                   |                                              |
|-----|-----------------------------------------------------------------------------------|----------------------------------------------|
| (a) | $\frac{T^o}{3} \text{thin}(Q) P(F(\frac{H}{3} \text{est}(R)) h) \text{est}(R)$    | $P(S) \text{est}(R) \sqsubseteq \exists S$   |
| (b) | $\sqsubseteq \frac{T^o}{3} \text{thin}(Q) \exists F(\frac{H}{3} \text{est}(R)) h$ | $\text{thin}(Q) \exists \sqsubseteq \exists$ |
| (c) | $\sqsubseteq \frac{T^o}{3} \exists F(\frac{H}{3} \text{est}(R)) h$                | $\frac{R}{3} \exists = R$                    |
| (d) | $= T^o F(\frac{H}{3} \text{est}(R)) h$                                            | $\frac{S}{3} \text{est}(R) \sqsubseteq S$    |
| (e) | $\sqsubseteq T^o F(H) h$                                                          | $T^o F(H) h = H$                             |
| (f) | $= H$                                                                             |                                              |

$$\text{Map}(h) \wedge F(R^\circ) \ h \sqsubseteq h \ R^\circ \wedge R^\circ \ R^\circ \sqsubseteq R^\circ \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) \ P(F(\frac{H}{\exists} \text{est}(R)) \ h) \ \text{est}(R) \sqsubseteq R^\circ \quad (9.1.1f)$$

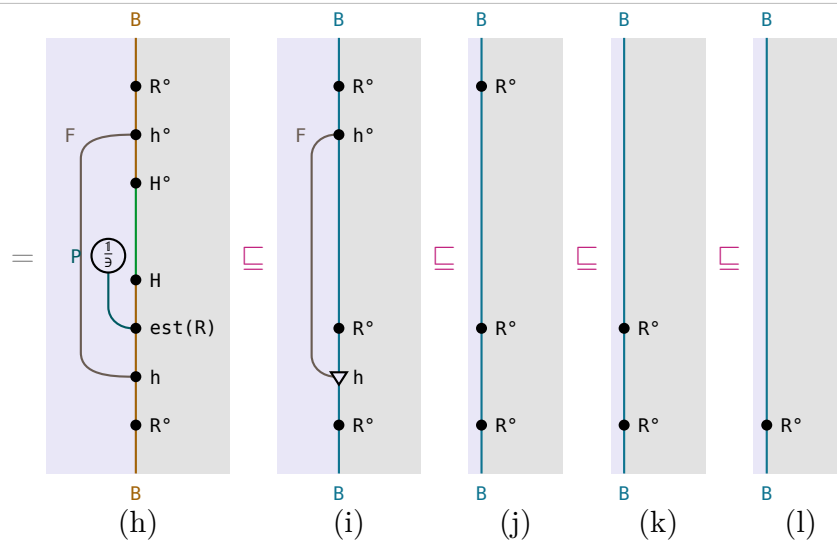
$H^\circ$  followed by the body at  $M$  is  $\mathbb{E}R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input



|     |                                                                                                                                                 |                                                                       |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| (a) | $H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) \ P(F(\frac{H}{\exists} \text{est}(R)) \ h) \ \text{est}(R)$                                    | $P(S) \ \text{est}(R) \sqsubseteq \mathbb{E} \setminus (S \ R^\circ)$ |
| (b) | $\sqsubseteq H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) \ (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) \ h \ R^\circ))$        | $H^\circ = h^\circ \ F(H^\circ) \ T$                                  |
| (c) | $= h^\circ \ F(H^\circ) \ T \frac{T^\circ}{\exists} \text{thin}(Q) \ (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) \ h \ R^\circ))$ | $T \frac{T^\circ}{\exists} \sqsubseteq \mathbb{E}$                    |
| (d) | $\sqsubseteq h^\circ \ F(H^\circ) \ \mathbb{E} \ \text{thin}(Q) \ (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) \ h \ R^\circ))$    | $\mathbb{E} \ \text{thin}(Q) \sqsubseteq Q^\circ \ \mathbb{E}$        |



|     |                                                                                                                                       |                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| (e) | $\sqsubseteq h^\circ \ F(H^\circ) \ Q^\circ \ \mathbb{E} \ (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) \ h \ R^\circ))$ | $S \ (S \setminus R) \sqsubseteq R$                                                     |
| (f) | $\sqsubseteq h^\circ \ F(H^\circ) \ Q^\circ \ F(\frac{H}{\exists} \text{est}(R)) \ h \ R^\circ$                                       | $\text{ThinCondition}(T, h, R, Q)$                                                      |
| (g) | $\sqsubseteq R^\circ \ h^\circ \ F(H^\circ) \ F(\frac{H}{\exists} \text{est}(R)) \ h \ R^\circ$                                       | $\Rightarrow h^\circ \ F(H^\circ) \ Q^\circ \sqsubseteq R^\circ \ h^\circ \ F(H^\circ)$ |

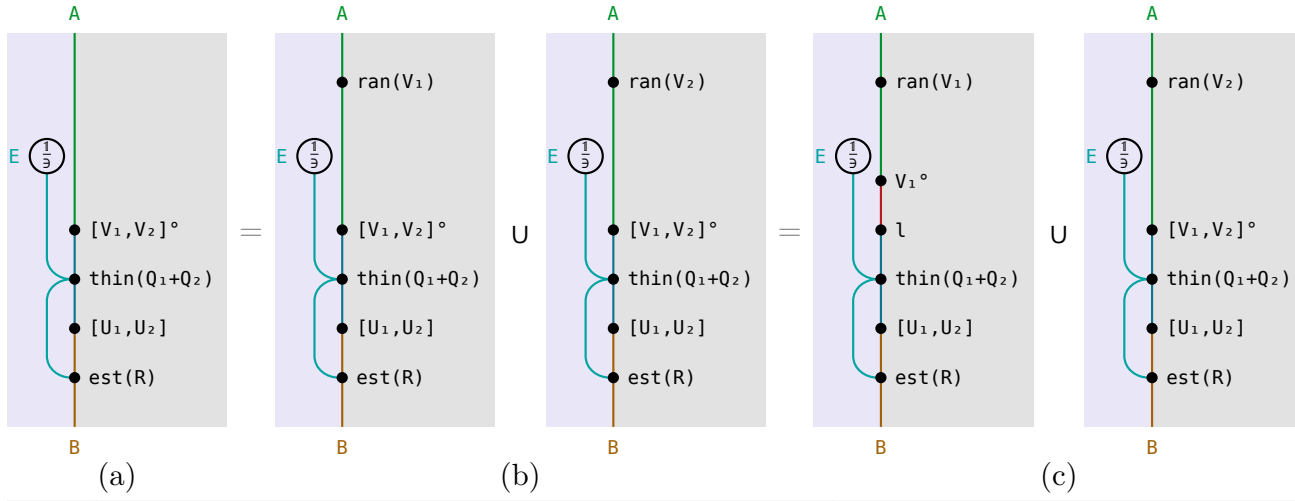


|     |                                                                               |                                                                                   |
|-----|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| (h) | $= R^\circ \ h^\circ \ F(H^\circ \ \frac{H}{\exists} \ est(R)) \ h \ R^\circ$ | $S^\circ \ \frac{S}{\exists} \ est(R) \subseteq R^\circ$                          |
| (i) | $\sqsubseteq R^\circ \ h^\circ \ F(R^\circ) \ h \ R^\circ$                    | Map(f)<br>$\implies F(R) \ f \subseteq f \ R \iff f^\circ \ F(R) \ f \subseteq R$ |
| (j) | $\sqsubseteq R^\circ \ R^\circ \ R^\circ$                                     | $R^\circ \ R^\circ \subseteq R^\circ$                                             |
| (k) | $\sqsubseteq R^\circ \ R^\circ$                                               | $R^\circ \ R^\circ \subseteq R^\circ$                                             |
| (l) | $\sqsubseteq R^\circ$                                                         |                                                                                   |

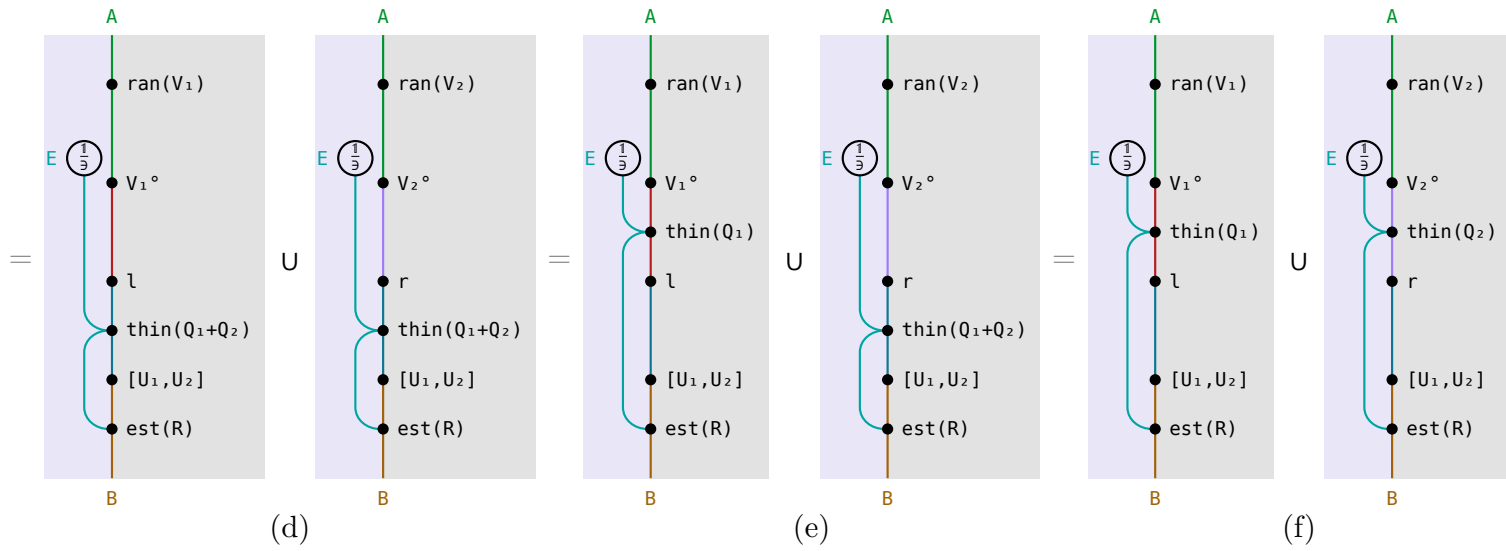
Proposition 9.1

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R) = (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(U_2) \ \text{est}(R))$$

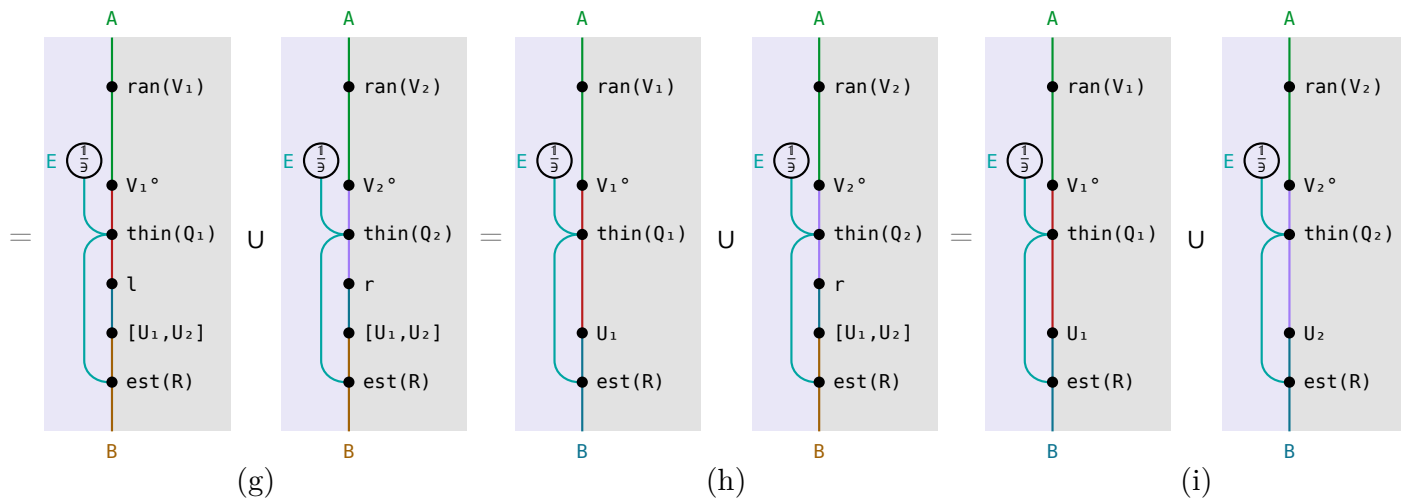
when  $V_1$  and  $V_2$  have disjoint ranges, the body over  $[V_1, V_2]$  runs the  $V_1$  problem on inputs  $V_1$  reaches and the  $V_2$  problem on inputs  $V_2$  reaches



|     |                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                          |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)$                                                                                                                                          | $\frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R) = (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$ |
| (b) | $= (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$   | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ P(l)$                                                                                                                                                                           |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ P(l) \ \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$ | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ P(r)$                                                                                                                                                                           |



|     |                                                                                                                                                                                                                                      |                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| (d) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ P(l) \ \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ P(r) \ \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$ | $P(l) \ \text{thin}(Q_1 + Q_2) = \text{thin}(Q_1) \ P(l)$ |
| (e) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ P(r) \ \text{thin}(Q_2) \ P([U_1, U_2]) \ \text{est}(R))$             | $P(r) \ \text{thin}(Q_1 + Q_2) = \text{thin}(Q_2) \ P(r)$ |
| (f) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$             |                                                           |



|     |                                                                                                                                                                                                                          |                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| (g) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$ | $l \ [R, S] = R$ |
| (h) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$               | $r \ [R, S] = S$ |
| (i) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ \text{thin}(Q_2) \ P(U_2) \ \text{est}(R))$                             |                  |

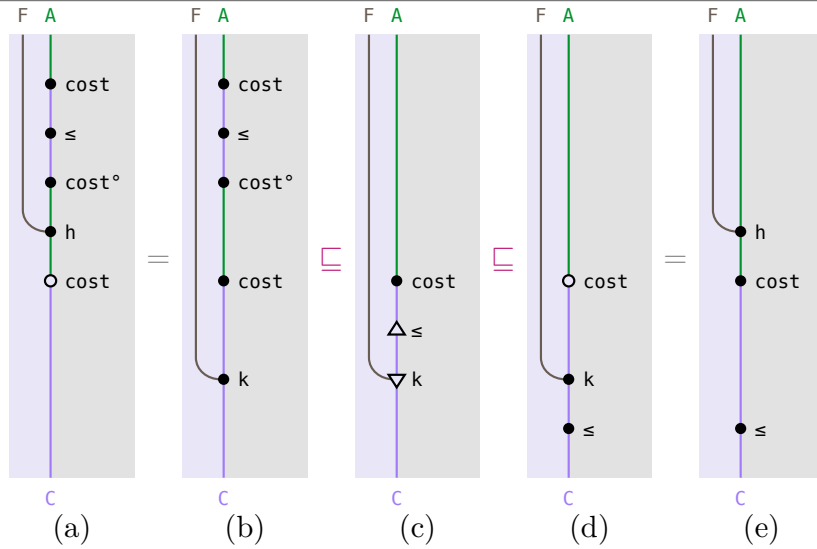


**Map(cost)  $\wedge$  h cost = F(cost) k  $\wedge$  F( $\leq$ ) k  $\sqsubseteq$  k  $\leq \implies$  F(cost  $\leq$  cost $^\circ$ ) h  $\sqsubseteq$  h cost  $\leq$  cost $^\circ$**  Proposition 9.2

if h then **cost** equals **F(cost)** then a **k** monotonic on  $\leq$ , then **h** is monotonic on **cost $\leq$ cost $^\circ$** ,  
which compares two values by comparing their **costs** under  $\leq$

**Map(cost)  $\wedge$  h cost = F(cost) k  $\wedge$  F( $\leq$ ) k  $\sqsubseteq$  k  $\leq \implies$  F(cost  $\leq$  cost $^\circ$ ) h cost  $\sqsubseteq$  h cost  $\leq$**

by shunting at the map **cost**, **F(cost $\leq$ cost $^\circ$ )h $\sqsubseteq$ h cost $\leq$ cost $^\circ$**  is **F(cost $\leq$ cost $^\circ$ )h cost $\sqsubseteq$ h cost  $\leq$** , which the chain proves



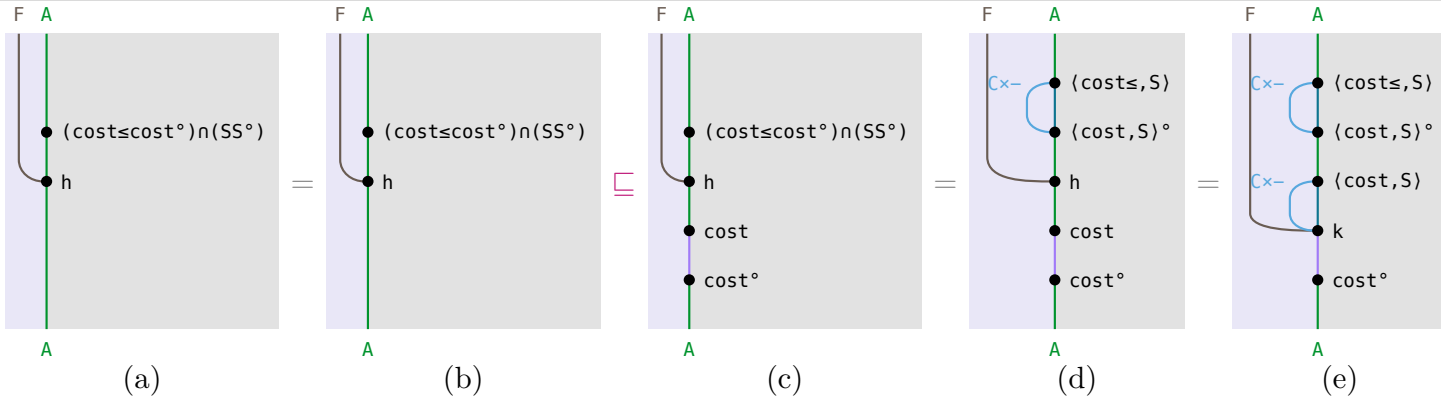
|     |                                                                    |                                                                                                  |
|-----|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| (a) | $F(\text{cost} \leq \text{cost}^\circ) \text{ h cost}$             | $\text{h cost} = F(\text{cost}) \text{ k}$                                                       |
| (b) | $= F(\text{cost} \leq \text{cost}^\circ) F(\text{cost}) \text{ k}$ | $\text{Map}(f) \wedge R = f \circ S \circ f^\circ$<br>$\implies R \circ f \sqsubseteq f \circ S$ |
| (c) | $\sqsubseteq F(\text{cost}) F(\leq) \text{ k}$                     | $F(\leq) \text{ k} \sqsubseteq \text{k} \leq$                                                    |
| (d) | $\sqsubseteq F(\text{cost}) \text{ k} \leq$                        | $\text{h cost} = F(\text{cost}) \text{ k}$                                                       |
| (e) | $= \text{h cost} \leq$                                             |                                                                                                  |

(9.1.1g)

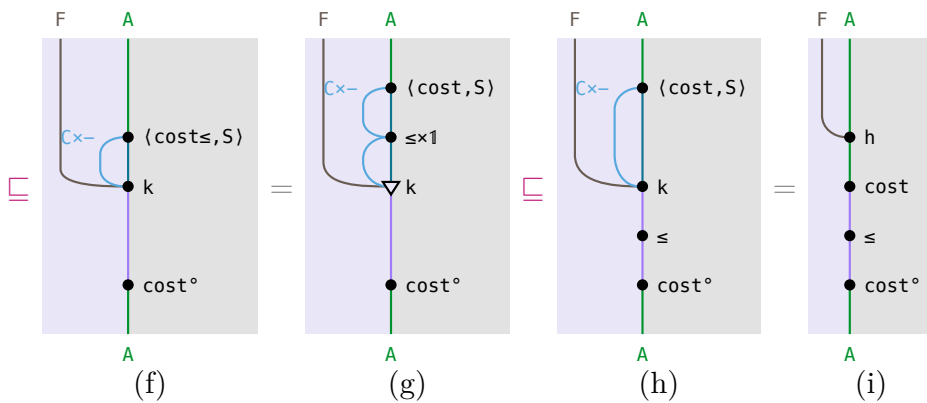
|   | the condition                                                                                                                                                                                                                                                                         | how it is discharged                                                                      |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1 | $F(R) \text{ h} \sqsubseteq \text{h} R$<br>Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $\text{h cost} = F(\text{cost}) \text{ k}$ , $F(\leq) \text{ k} \sqsubseteq \text{k} \leq$                                                                            | monotonicity when the cost is itself a fold with a step <b>k</b> monotonic on $\leq$      |
| 2 | $F(R \circ (H^\circ H)) \text{ h} \sqsubseteq \text{h} R$<br>Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $\text{h cost} = F(\langle \text{cost}, H^\circ \rangle) \text{ k}$ ,<br>$F(\leq \times 1) \text{ k} \sqsubseteq \text{k} \leq$ , $H^\circ$ simple; | monotonicity <b>in context</b> : <b>k</b> may also read the input the part was built from |
| 3 | $QF(H) \text{ h} \sqsubseteq F(H) \text{ h} R$ at $Q \triangleq F(U, V)$<br>Proposition 9.4, $U, V$ preorders, $F(U, R) \text{ h} \sqsubseteq \text{h} R$ , $V H \sqsubseteq H R$                                                                                                     | both conditions at once, split along the two arguments of a bifunctor                     |

**Map(cost)  $\wedge$  Simple(S)  $\wedge$  h cost = F( $\langle$ cost,S $\rangle$ ) k  $\wedge$  F( $\leq \times 1$ ) k  $\sqsubseteq$  k  $\leq \Rightarrow$  F( $\langle$ cost  $\leq$  cost $^\circ$ )  $\cap$  (S S $^\circ$ ) h  $\sqsubseteq$  h cost  $\leq$  cost $^\circ$**  Proposition 9.3

improving each part by **R** within its **S**-context, then assembling by **h**, is below **hR**



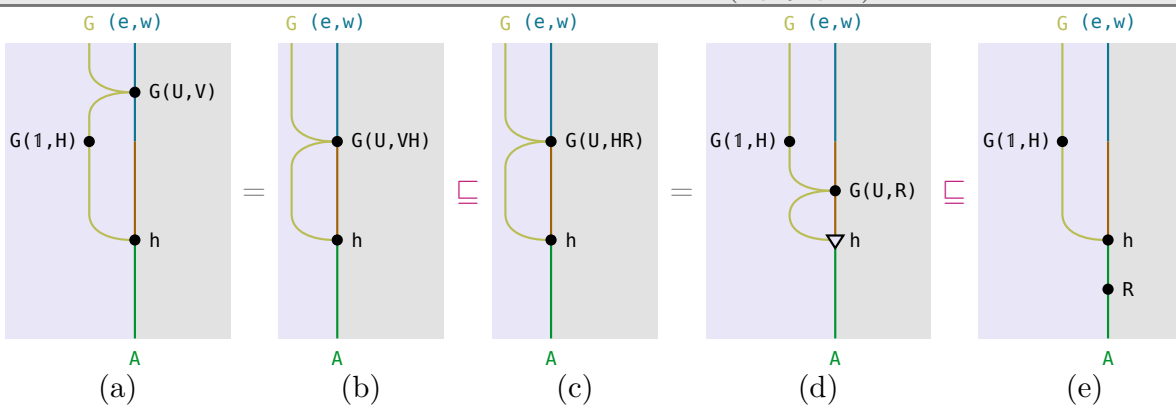
|     |                                                                                                                                     |                                                                                  |
|-----|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| (a) | $F((\text{cost} \leq \text{cost}^\circ) \cap (S S^\circ)) h$                                                                        |                                                                                  |
| (b) | $= F((\text{cost} \leq \text{cost}^\circ) \cap (S S^\circ)) h$                                                                      | $\text{Map}(f)$<br>$\Rightarrow 1 \sqsubseteq f f^\circ$                         |
| (c) | $\sqsubseteq F((\text{cost} \leq \text{cost}^\circ) \cap (S S^\circ)) h \text{ cost cost}^\circ$                                    | $\langle X, Y \rangle \langle R, S \rangle^\circ = (X R^\circ) \cap (Y S^\circ)$ |
| (d) | $= F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ) h \text{ cost cost}^\circ$                           | $h \text{ cost} = F(\langle \text{cost}, S \rangle) k$                           |
| (e) | $= F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ \langle \text{cost}, S \rangle) k \text{ cost}^\circ$ |                                                                                  |



|     |                                                                             |                                                              |
|-----|-----------------------------------------------------------------------------|--------------------------------------------------------------|
| (f) | $\sqsubseteq F(\langle \text{cost} \leq, S \rangle) k \text{ cost}^\circ$   | $\langle X, Y \rangle (R \times 1) = \langle X R, Y \rangle$ |
| (g) | $= F(\langle \text{cost}, S \rangle) F(\leq \times 1) k \text{ cost}^\circ$ | $F(\leq \times 1) k \sqsubseteq k \leq$                      |
| (h) | $\sqsubseteq F(\langle \text{cost}, S \rangle) k \leq \text{cost}^\circ$    | $h \text{ cost} = F(\langle \text{cost}, S \rangle) k$       |
| (i) | $= h \text{ cost} \leq \text{cost}^\circ$                                   |                                                              |

**G(U,R) h  $\sqsubseteq$  h R  $\wedge$  V H  $\sqsubseteq$  H R  $\Rightarrow$  G(U,V) G(1,H) h  $\sqsubseteq$  G(1,H) h R** Proposition 9.4

thinning the parts by **U** in the first argument and by **V** in the second (**G(U,V)**), then solving by **H** and assembling by **h**, gives only what solving and assembling and then improving by **R** gives (**G(1,H)hR**)



|     |                          |                            |
|-----|--------------------------|----------------------------|
| (a) | $G(U,V) G(1,H) h$        |                            |
| (b) | $= G(U,VH) h$            | $V H \sqsubseteq H R$      |
| (c) | $\sqsubseteq G(U,HR) h$  |                            |
| (d) | $= G(1,H) G(U,R) h$      | $G(U,R) h \sqsubseteq h R$ |
| (e) | $\sqsubseteq G(1,H) h R$ |                            |

## §9.2. The string edit problem

(9.2a)

|    | name               | type                                                                            | definition                                                                                                                                                                                                                                                                                                                                                                        | meaning                                                                                                                     |
|----|--------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>Op</b>          | Type                                                                            | $(\exists a. o = \text{cpy}(a)) \vee (\exists a. o = \text{del}(a)) \vee (\exists a. o = \text{ins}(a))$                                                                                                                                                                                                                                                                          | an edit operation copies, deletes or inserts one character                                                                  |
| 2  |                    |                                                                                 | $FX = 1 + (Op \times X)$                                                                                                                                                                                                                                                                                                                                                          | $F(Op, X)$ is the empty sequence or one operation paired with an $X$                                                        |
| 3  | <b>edit</b>        | $[Op] \rightarrow [Char] \times [Char]$                                         | $\text{edit} = ([\text{base}, \text{step}])$ , $\text{edit}([]) = ([], [])$ , $\text{edit}([op] \# es) = \text{step}(r((op, \text{edit}(es))))$                                                                                                                                                                                                                                   | the two strings an edit sequence turns one into the other                                                                   |
| 4  |                    |                                                                                 | $\text{length} = ([\text{zero}, \pi_2 \text{ succ}])$                                                                                                                                                                                                                                                                                                                             | the number of operations in a sequence                                                                                      |
| 5  | <b>R</b>           | $[Op] \rightarrow [Op]$                                                         | $R = \text{length} \leq \text{length}^\circ$ , $R(es, fs) \iff \text{length}(es) \leq \text{length}(fs)$                                                                                                                                                                                                                                                                          | $es$ is no longer than $fs$                                                                                                 |
| 6  | <b>V</b>           | $[Char] \times [Char] \rightarrow [Char] \times [Char]$                         | $V = \text{suffix}^\circ \times \text{suffix}^\circ$ , $V(p, q) \iff \text{suffix}(\pi_1(p), \pi_1(q)) \wedge \text{suffix}(\pi_2(p), \pi_2(q))$                                                                                                                                                                                                                                  | each string of the first pair is a suffix of the matching string of the second                                              |
| 7  | <b>Q</b>           | $F([Char] \times [Char]) \rightarrow F([Char] \times [Char])$                   | $Q(l(val), l(val')) = \text{true}$ , $Q(r(p), r(q)) = V(\pi_2(p), \pi_2(q))$ , $(\forall val. \forall val'. u = l(val) \implies v = l(val') \implies \text{false}) \wedge (\forall p. \forall q. u = r(p) \implies v = r(q) \implies \text{false}) \implies Q(u, v) = \text{false}$ , $Q(l(u), l(u))$ , $Q(r(p), r(q)) \iff V(\pi_2(p), \pi_2(q))$                                | the identity on the <b>base</b> summand; on the <b>step</b> summand any two operations, and the two pairs of strings by $V$ |
| 8  | <b>empty</b>       | $[Char] \times [Char] \rightarrow [Char] \times [Char]$                         | $\text{empty}(p, q) \iff p = q \wedge p = ([], [])$                                                                                                                                                                                                                                                                                                                               | holds only at the pair of two empty strings                                                                                 |
| 9  | <b>unstep</b>      | $[Char] \times [Char] \rightarrow [Op] \times [Char] \times [Char]$             | $q \in \text{unstep}(p) \implies \text{step}(r(q)) = p$ , $\text{unstep}([], []) = []$ , $\text{unstep}([a] \# xs, []) = [(\text{del}(a), (xs, []))]$ , $\text{unstep}([], [b] \# ys) = [(ins(b), ([], ys))]$ , $\text{unstep}([a] \# xs, [b] \# ys) = \text{if } a = b \text{ then } [(cpy\ a, xs, ys)] \text{ else } [(\text{del}\ a, xs, [b] \# ys), (ins\ b, [a] \# xs, ys)]$ | every split of a pair of strings into one operation and a smaller pair; <b>step</b> puts it back                            |
| 10 | <b>[nil, cons]</b> | $L + E \times \mu F \rightarrow \mu F$                                          | $[\text{nil}, \text{cons}](l(u)) = []$ , $[\text{nil}, \text{cons}](r(o, es)) = [o] \# es$                                                                                                                                                                                                                                                                                        | $\alpha = [\text{nil}, \text{cons}]$ : the empty edit sequence, or one operation in front of a sequence                     |
| 11 | <b>step</b>        | $\text{Unit} + Op \times [Char] \times [Char] \rightarrow [Char] \times [Char]$ | $\text{step}(l(u)) = ([], [])$ , $\text{step}(r((cpy(a), (xs, ys)))) = ([a] \# xs, [a] \# ys)$ , $\text{step}(r((\text{del}(a), (xs, ys)))) = ([a] \# xs, ys)$ , $\text{step}(r((ins(a), (xs, ys)))) = (xs, [a] \# ys)$                                                                                                                                                           | $[\text{base}, \text{step}]$ : the two empty strings, or one operation added to the front of the pair                       |

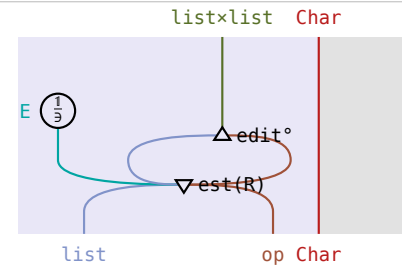
$$\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle}) \text{cons}) \text{minlist}(R)), (\mu X : \frac{[\text{base}, \text{step}]^\circ}{3} \text{thin}(Q) \text{P}(\text{F}(X) [\text{nil}, \text{cons}]) \text{est}(R)) \sqsubseteq \frac{\text{edit}^\circ}{3} \text{est}(R) \quad (9.2b)$$

a shortest edit sequence from which both strings can be reconstituted is one pass over the two of them, each step copying, deleting or inserting one character and the best sequence for what is left taken from the entries already computed

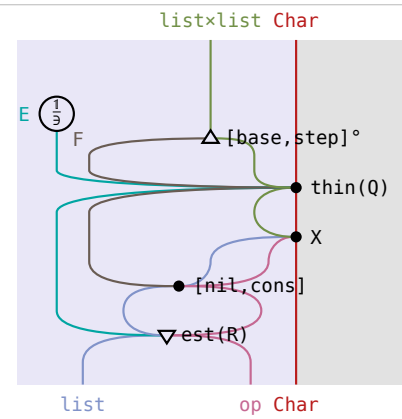
## circuit

$[\text{Char}] \xrightarrow{\frac{1}{3}} \text{P}[\text{Char}]^2 \xrightarrow{\text{E}(\text{edit}^\circ)} \text{P}[0p] \xrightarrow{\text{est}(R)} [0p]$   
 the specification —  $\text{edit}([\text{ }]) = ([\text{ }], [\text{ }])$ ,  $\text{edit}([op] \# es) = \text{step}(r((op, \text{edit}(es))))$

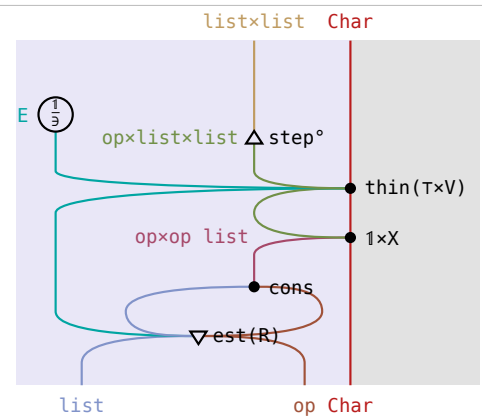
## Hinze–Marsden



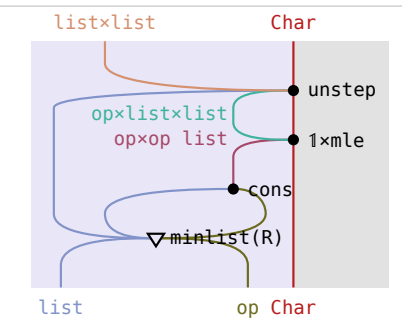
$[\text{Char}] \xrightarrow{\frac{1}{3}} \text{P}[\text{Char}]^2 \xrightarrow{\text{E}([\text{base}, \text{step}]^\circ)} \text{thin}(Q) \text{P}[\text{Char}]^2 \xrightarrow{\text{P}(\text{F}(X) [\text{nil}, \text{cons}])} \text{P}[0p] \xrightarrow{\text{est}(R)} [0p]$   
 at  $Q \triangleq 1 + (U \times V)$ . Monotonicity  $\text{F}(R) \alpha \sqsubseteq \alpha R$  is Proposition 9.2 at  $\text{length} \triangleq ([\text{zero}, \pi_2 \text{succ}])$  with  $\text{succ}$  monotonic on  $\leq$ , so  $\text{cons}$  is monotonic on  $R$ . The thinning condition  $\sqsubseteq QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$  over  $\text{F}(0p, [\text{Char}] \times [\text{Char}])$  is Proposition 9.4 at  $U \triangleq T$  —  $\text{F}(T, R) \alpha \sqsubseteq \alpha R$ , left as an exercise, so any two operations may be compared — and  $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ , —  $\text{edit}(1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}$ ,  $\text{edit}(\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$ .  $\text{edit} \cdot \text{suffix} = \text{tail}^*$  and  $BA \sqsubseteq CB \Rightarrow BA^* \sqsubseteq C^*B$  cut those to one step each: drop the operation that produced the head, or weaken its  $\text{cpy}$  to a  $\text{del}$ , never lengthening the sequence



$[\text{Char}] \xrightarrow{\frac{1}{3}} \text{P}[\text{Char}]^2 \xrightarrow{\text{E}(\text{step}^\circ)} \text{thin}(T \times V) \text{P}(0p \times [\text{Char}] \times [\text{Char}]) \xrightarrow{\text{P}((1 \times X) \text{cons})} \text{P}[0p] \xrightarrow{\text{est}(R)} [0p]$   
 = Proposition 9.1:  $\text{base}$  and  $\text{step}$  have disjoint ranges, so the fixed point splits into one branch per summand —  $\text{empty}$ , the coreflexive on  $(xs, ys)$  with both lists empty, is where  $\text{base}$  returns



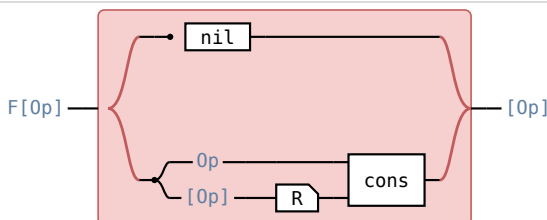
$[\text{Char}] \xrightarrow{\text{unstep}} [0p \times [\text{Char}] \times [\text{Char}]] \xrightarrow{\text{list}((1 \times \text{mle}) \text{cons})} [[0p]] \xrightarrow{\text{minlist}(R)} [0p]$   
 $\text{unstep}$  implements  $\frac{\text{step}^\circ}{3} \text{thin}(U \times V)$  — at most two decompositions survive, a copy beating a delete or an insert wherever it is available — and  $\text{minlist}(R)$  implements  $\text{est}(R)$ . The same subproblem is solved many times over, so the running time is exponential in the two lengths



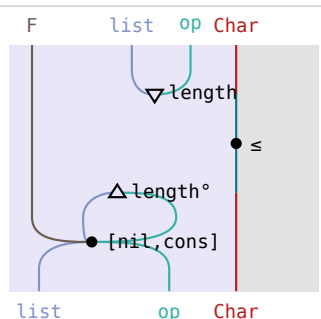
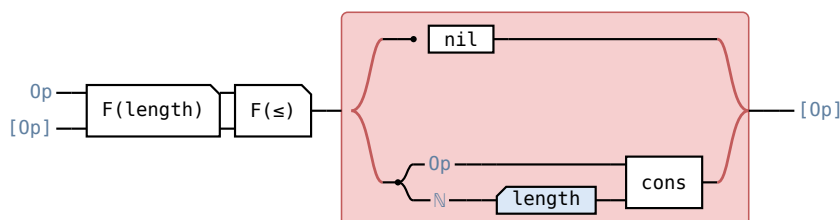
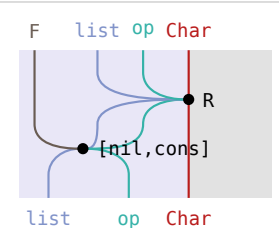
$$\text{F}(R) [\text{nil}, \text{cons}] \sqsubseteq [\text{nil}, \text{cons}] R \quad (9.2c)$$

comparing the tails by  $R$  and then putting an operation in front ( $\text{F}(R) \alpha$ ) relates only sequences that  $R$  also relates once the operation is in front ( $\alpha R$ )

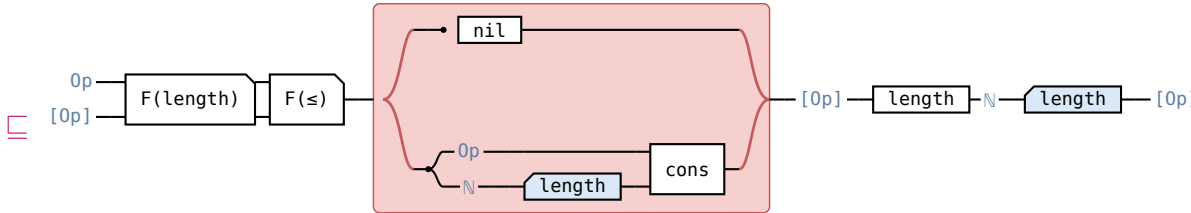
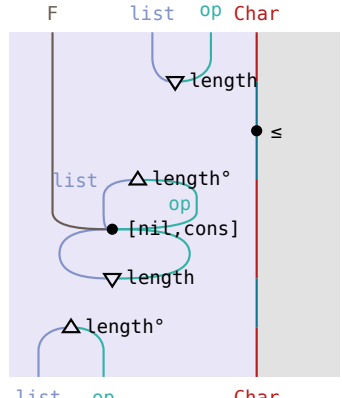
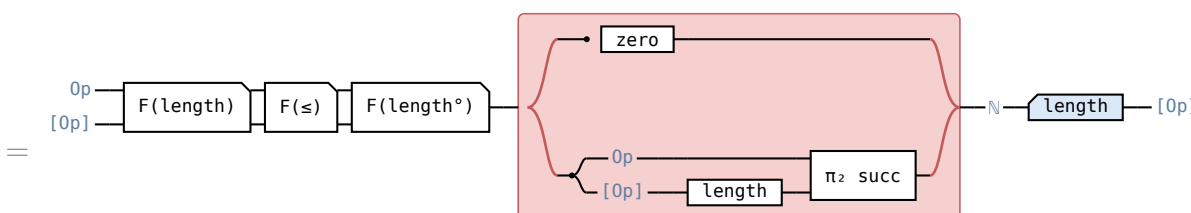
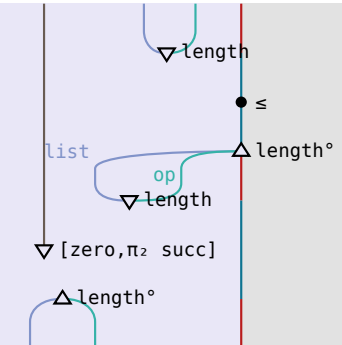
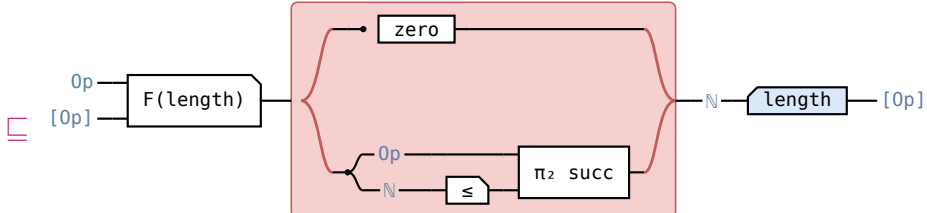
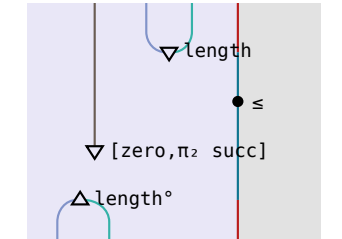
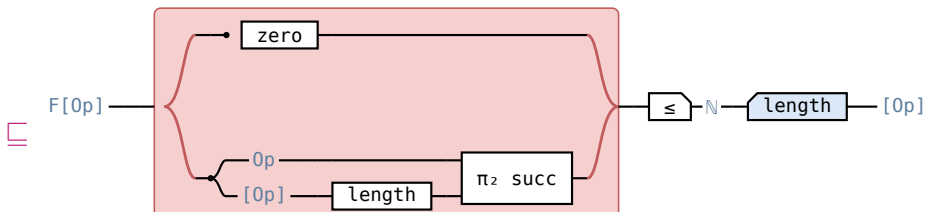
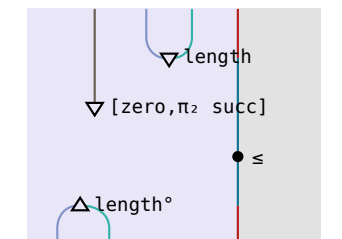
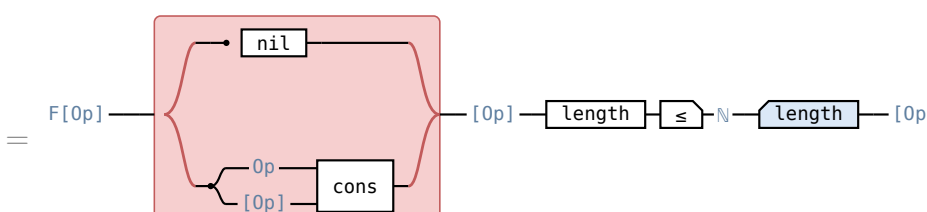
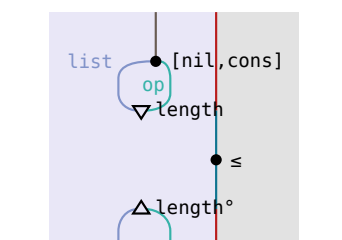
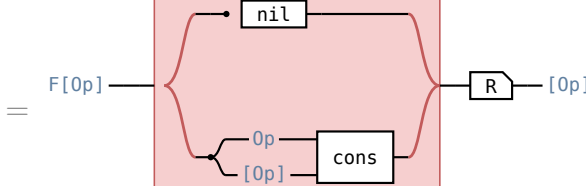
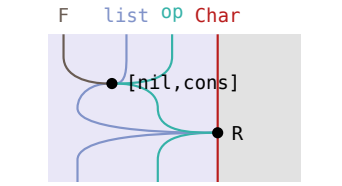
## circuit



## Hinze–Marsden



$R \triangleq \text{length} \leq \text{length}^\circ$  —  $es \ R \ fs \iff \text{length}(es) \leq \text{length}(fs)$ ;  $F$  preserves composition

| circuit                                                                                                                                                                                                                                                                                                                            | Hinze–Marsden                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <p><math>\text{length}</math> is a map, so entire: <math>1 \sqsubseteq \text{length} \text{ length}^\circ</math></p>                                                                                                                            |    |
|  <p><math>\alpha \text{ length} = F(\text{length})[\text{zero}, \pi_2 \text{ succ}] : \text{length}^\Delta([ \text{zero}, \pi_2 \text{ succ} ]) - \text{es } R \text{ fs} \iff \text{length}(\text{es}) \leq \text{length}(\text{fs})</math></p> |    |
|  <p><math>\text{length}</math> is a map, so simple, and <math>F</math> is monotonic: <math>F(\text{length}^\circ)F(\text{length}) \sqsubseteq F(1) = 1</math></p>                                                                              |  |
|  <p><math>\text{succ}</math> is monotonic on <math>\leq</math>: <math>F(\leq)[\text{zero}, \pi_2 \text{ succ}] \sqsubseteq [\text{zero}, \pi_2 \text{ succ}] \leq</math></p>                                                                   |  |
|  <p><math>F(\text{length})[\text{zero}, \pi_2 \text{ succ}] = \alpha \text{ length}</math> again</p>                                                                                                                                           |  |
|  <p><math>R \sqsubseteq \text{length} \leq \text{length}^\circ - \text{es } R \text{ fs} \iff \text{length}(\text{es}) \leq \text{length}(\text{fs})</math></p>                                                                                 |  |

| $Q \text{ F(edit}^\circ) [\text{nil}, \text{cons}] \sqsubseteq \text{F(edit}^\circ) [\text{nil}, \text{cons}] R$ (9.2d)                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| every edit sequence built from a decomposition $Q$ puts above a given one — any operation, each string lengthened at the front — is at least as long as one built from the given one                                                                                                                                                                                                                                                                                                                                                                                                         |               |
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |
| $=$ <p> <math>Q \triangleq 1 + (U \times V)</math> is <math>F(U, V)</math> at <math>U \triangleq \tau - Q(l(val), l(val')) = \text{true}</math>, <math>Q(r(p), r(q)) = V(\pi_2(p), \pi_2(q))</math>, <math>(\forall val. \forall val'. u = l(val) \Rightarrow v = l(val') \Rightarrow \text{false}) \wedge (\forall p. \forall q. u = r(p) \Rightarrow v = r(q) \Rightarrow \text{false})</math><br/> <math>\Rightarrow Q(u, v) = \text{false}</math>; and the bifunctor <math>F</math> preserves composition: <math>F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)</math> </p> |               |
| $\sqsubseteq$ <p>Proposition 9.4's second condition <math>V \text{ edit}^\circ \sqsubseteq \text{edit}^\circ R - \text{edit } V^\circ \sqsubseteq R^\circ \text{ edit}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                            |               |
| $=$ <p><math>F</math> preserves composition: <math>F(U, \text{edit}^\circ R) = F(1, \text{edit}^\circ)F(U, R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |
| $\sqsubseteq$ <p>Proposition 9.4's first condition <math>F(\tau, R) \alpha \alpha R</math>, left as an exercise in the book: <b>cons</b> adds one to both lengths whatever the two operations are</p>                                                                                                                                                                                                                                                                                                                                                                                        |               |

| $\text{edit } V^\circ \sqsubseteq R^\circ \text{ edit}$ (9.2e)<br>cutting a front off either string an edit sequence produces ( $\text{edit } V^\circ$ ) leaves a pair that a sequence no longer produces ( $R^\circ \text{ edit}$ )                                                                                                                  |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                               | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                                                                       |               |
| $=$<br>$V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ , and $\times$ preserves composition                                                                                                                                                                                                                                             |               |
| $\sqsubseteq$<br>$\text{edit } (\text{suffix} \times 1) \sqsubseteq R^\circ \text{ edit}$ : drop the operation that produced the head, or weaken its <b>cpy</b> to an <b>ins</b> . Proved by induction at <b>suffix</b> directly, not through $\text{suffix} = \text{tail}^*$ and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$ as the book does |               |
| $\sqsubseteq$<br>$\text{edit } (1 \times \text{suffix}) \sqsubseteq R^\circ \text{ edit}$ , the mirror image: <b>cpy</b> weakens to a <b>del</b>                                                                                                                                                                                                      |               |
| $\sqsubseteq$<br>$R^\circ$ is transitive, $\leq$ being so                                                                                                                                                                                                                                                                                             |               |

| step base° = 0 (9.2f)                                                                                     |                             |
|-----------------------------------------------------------------------------------------------------------|-----------------------------|
| <p><b>circuit</b></p>                                                                                     | <p><b>Hinze–Marsden</b></p> |
| <p><b>=</b></p> <p>base returns only ([], []), so base°=empty base°</p>                                   |                             |
| <p><b>=</b></p> <p>step empty=0: cpy and del put a character on the left string, ins one on the right</p> |                             |
| <p><b>=</b></p> <p>0 composed with anything is 0</p>                                                      |                             |

(9.2g)

```

1  mle(xs,ys) = head(column(xs,ys)) , column(xs,ys)=[mle(u,ys)|u<-tails(xs)]
   = column(xs)=( $\llbracket$ [fstcol(xs),nextcol(xs)] $\rrbracket$ ), fstcol=list(del) tails
   the tabulation: mle(xs,ys) needs mle(u,v) for every tail u of xs and v of ys, so the columns are built right to left

2  column(xs,[b]*ys) = nextcol(xs,(b,column(xs,ys)))
   = nextcol(xs)(b,us)=( $\llbracket$ [base(b,last(us)),step(b)] $\rrbracket$ (xus), xus=zip(xs,zip(init(us),tail(us)))
   each column is a fold built bottom to top, over xs zipped with the adjacent pairs of the column to its right

3  base(b,u)=[ins(b)]#u
   = step(b)((a,(u,v)),ws)=(a=b $\rightarrow$ [cpy(a)]#v)#ws, [bmin(R)([del(a)]#w,[ins(b)]#u)]#ws
   w=head(ws); these base, step are not edit's. An entry depends on the one below it (a delete), the one to its right (an insert), and the one
   below that (a copy) — quadratic in the two lengths

```

### §9.3. Optimal bracketing



(9.3a)

|    | name                                       | type                                                                 | definition                                                                                                                                                                                                                                       | meaning                                                                                                    |
|----|--------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1  | tree(A)                                    | Type                                                                 | $\text{tree}(A) \triangleq \text{tip}(A) \mid \text{bin}(\text{tree}(A), \text{tree}(A))$                                                                                                                                                        | a tree is a tip holding one value, or a node joining two trees; $h \triangleq [\text{tip}, \text{bin}]$    |
| 2  | flatten                                    | $\text{tree}(A) \rightarrow \text{LA}$                               | $\langle [\text{wrap}, \text{cat}] \rangle = \text{flatten}, \text{flatten}(\text{tip}(a)) = [a], \text{flatten}(\text{bin}(l, r)) = \text{cat}(\text{flatten}(l), \text{flatten}(r))$                                                           | the tips of a tree, left to right; $H = \text{flatten}^\circ$ builds every tree over a list                |
| 3  | opb(p)                                     | $\text{Int} \times \text{S}$                                         | $\text{opb}(p) \triangleq ((\text{cb}(\pi_2(\pi_1(p)), \pi_2(\pi_2(p))) + \pi_1(\pi_1(p))) + \pi_1(\pi_2(p)), \text{sb}(\pi_2(\pi_1(p)), \pi_2(\pi_2(p))))$                                                                                      | joining two subtrees costs <b>cb</b> of their sizes plus the cost of each                                  |
| 4  | $\langle \text{cost}, \text{size} \rangle$ | $\text{tree}(A) \rightarrow \text{Int} \times \text{S}$              | $\langle \text{cost}, \text{size} \rangle(\text{tip}(a)) = (0, \text{st}(a)), \langle \text{cost}, \text{size} \rangle(\text{bin}(l, r)) = \text{opb}(\langle \text{cost}, \text{size} \rangle(l), \langle \text{cost}, \text{size} \rangle(r))$ | the cost of evaluating a bracketing, paired with the size of its result                                    |
| 5  | sz                                         | $\text{LA} \rightarrow \text{S}$                                     | $\text{sz}([a]) = \text{st}(a), \text{sz}([a] \# x) = \text{sb}(\text{st}(a), \text{sz}(x)), \text{Assoc}(\text{sb}) \implies \pi_2(\langle \text{cost}, \text{size} \rangle(t)) = \text{sz}(\text{flatten}(t))$                                 | <b>sb</b> associative, so the size depends only on the flattened list                                      |
| 6  | R                                          | $\text{tree}(A) \rightarrow \text{tree}(A)$                          | $R = \text{cost} \leq \text{cost}^\circ, R(t, t') \iff \text{cost}(t) \leq \text{cost}(t')$                                                                                                                                                      | <b>t</b> costs no more than <b>t'</b>                                                                      |
| 7  | g                                          | $F(\text{Int} \times \text{LA}) \rightarrow \text{Int}$              | $g \triangleq [\neg 0, (1 \times \text{sz})^2 \text{ opb } \pi_1], g(l(a)) = 0, g(r((cx, x), cy, y)) = (\text{cb}(\text{sz}(x), \text{sz}(y)) + cx) + cy$                                                                                        | the cost of the top step alone: zero at a tip, <b>cb</b> of the two sizes at a node                        |
| 8  | nonsingle                                  | $\text{LA} \rightarrow \text{LA}$                                    | $x \text{ nonsingle } y \iff x = y \wedge (\exists a. \exists z. x = [a] \# z)$                                                                                                                                                                  | holds at the lists of two or more elements                                                                 |
| 9  | splits                                     | $\text{LA} \rightarrow [\text{L A} \times \text{L A}]$               | $\text{splits} = (\text{inits}^+, \text{tails}^+) \text{ zip}, \text{splits}(x) = \text{zip}(((\text{inits}^+)(x)), ((\text{tails}^+)(x)))$                                                                                                      | every way to cut a list into two non-empty parts — an implementation of $\frac{\text{cat}^\circ}{\exists}$ |
| 10 | array                                      | $\text{LA} \rightarrow [\text{tree A}]$                              | $\text{array} \triangleq \text{inits list}(\text{row}), \text{array}(x, \text{xss}) \iff \text{xss} = \text{list}(\text{row}, \text{inits}(x))$                                                                                                  | the best trees of every segment, held as rows                                                              |
| 11 | row                                        | $\text{LA} \rightarrow [\text{tree A}]$                              | $\text{row} = \text{tails list}(\text{mct}), \text{row}(x, \text{ys}) \iff \text{ys} = \text{list}(\text{mct}, \text{tails}(x))$                                                                                                                 | the best tree of every non-empty suffix                                                                    |
| 12 | col                                        | $\text{LA} \rightarrow [\text{tree A}]$                              | $\text{col} = \text{inits list}(\text{mct}), \text{col}(x, \text{ys}) \iff \text{ys} = \text{list}(\text{mct}, \text{inits}(x))$                                                                                                                 | the best tree of every non-empty prefix                                                                    |
| 13 | mix                                        | $[\text{tree A}] \times [\text{tree A}] \rightarrow \text{tree}(A)$  | $\text{mix} = \text{zip list}(\text{bin}) \text{ minlist}(R), \text{mix}(p, t) \iff t = \text{minlist}(R) (\text{list bin zip})$                                                                                                                 | join the matching prefix and suffix trees and keep a cheapest                                              |
| 14 | next                                       | $[\text{tree A}] \times [\text{tree A}] \rightarrow [\text{tree A}]$ | $\text{next} = (\pi_1, \text{mix}) \text{ snoc}, \text{next}(p, \text{zs}) \iff \text{zs} = \text{snoc}((\pi_1(p), \text{minlist}(R) (\text{list bin zip})))$                                                                                    | add the next best tree to the end of the column                                                            |
| 15 | process                                    | $\text{A} \times [\text{tree A}] \rightarrow [\text{tree A}]$        | $\text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next}), \text{process}((a, \text{xss}), \text{ys}) \iff \text{loop}(\text{next})(([\text{tip } a], \text{xss}), \text{ys})$                                           | start the column at a tip and run <b>next</b> over the rows                                                |

| $\text{Assoc}(\text{sb}) \Rightarrow F(\langle \text{cost}, \text{flatten} \rangle) \text{ g} = [\text{tip}, \text{bin}] \text{ cost}$ (9.5)<br>pairing each subtree with its cost and its flattening and then applying <b>g</b> gives what building the tree by <b>[tip, bin]</b><br>and taking its <b>cost</b> gives                                                                                                                                                                                                                                                                                                                                                                      |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Hinze–Marsden |
| $F(\text{tree}(A)) \rightarrow \boxed{F(\langle \text{cost}, \text{flatten} \rangle)} \rightarrow F(\text{Int} \times \text{LA}) \rightarrow \boxed{\text{g}} \rightarrow \text{Int}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \boxed{\text{l}} \rightarrow A \rightarrow \text{zero} \\ \boxed{\text{r}} \rightarrow \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{flatten} \rightarrow \text{LA} \rightarrow \text{sz} \end{array} \right\} \\ \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{flatten} \rightarrow \text{LA} \rightarrow \text{sz} \end{array} \right\} \end{array} \right\} \rightarrow \boxed{\text{opb}} \rightarrow \text{Int}$ <p>definition of <b>g</b>; coproducts and products — <math>\text{g} \triangleq [\rightarrow 0, (1 \times \text{sz})^2 \text{ opb } \pi_1]</math></p> |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \boxed{\text{l}} \rightarrow A \rightarrow \text{zero} \\ \boxed{\text{r}} \rightarrow \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{size} \end{array} \right\} \\ \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{size} \end{array} \right\} \end{array} \right\} \rightarrow \boxed{\text{opb}} \rightarrow \text{Int}$ <p><math>\text{flatten sz} = \text{size}</math>, since <b>sb</b> is associative — <math>\text{sz}([a]) = \text{st}(a)</math>, <math>\text{sz}([a] * x) = \text{sb}(\text{st}(a), \text{sz}(x))</math></p>                                          |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \boxed{\text{l}} \rightarrow A \rightarrow \text{zero} \\ \boxed{\text{r}} \rightarrow \text{tree}(A) \rightarrow \boxed{\text{bin}} \rightarrow \text{tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \rightarrow \text{Int} \\ \text{size} \rightarrow S \end{array} \right\} \end{array} \right\} \rightarrow \text{Int}$ <p><math>\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])</math>, at a node <math>\text{bin}(\text{cost}, \text{size}) = (\text{cost}, \text{size})^2 \text{ opb}</math></p>                                                                                |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \boxed{\text{l}} \rightarrow A \rightarrow \boxed{\text{tip}} \rightarrow \text{tree}(A) \rightarrow \text{cost} \\ \boxed{\text{r}} \rightarrow \text{tree}(A) \rightarrow \boxed{\text{bin}} \rightarrow \text{tree}(A) \rightarrow \text{cost} \end{array} \right\} \rightarrow \text{Int}$ <p><math>\text{tip cost} = \text{zero}</math>; products, <math>\langle \text{cost}, \text{size} \rangle \pi_1 = \text{cost}</math></p>                                                                                                                                                                                            |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \boxed{\text{l}} \rightarrow A \rightarrow \boxed{\text{tip}} \\ \boxed{\text{r}} \rightarrow \text{tree}(A) \rightarrow \boxed{\text{bin}} \end{array} \right\} \rightarrow \text{tree}(A) \rightarrow \boxed{\text{cost}} \rightarrow \text{Int}$ <p>coproducts, <math>[\text{tip cost}, \text{bin cost}] = [\text{tip}, \text{bin}] \text{ cost}</math></p>                                                                                                                                                                                                                                                                   |               |

| $F(\leq \times 1) \ g \sqsubseteq g \leq$ <span style="float: right;">(9.6)</span>                                                                                                                              |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| raising the cost of either subtree ( $\leq \times 1$ in both slots) and then applying $g$ returns a value at least what $g$ returns on the costs before raising ( $g \leq$ )                                    |               |
| circuit                                                                                                                                                                                                         | Hinze–Marsden |
| $F(\text{Int} \times \text{LA}) \rightarrow \boxed{F(\leq \times 1)} \rightarrow \boxed{g} \rightarrow \text{Int}$                                                                                              |               |
| $A + \text{Int} \times \text{LA} \times \text{Int} \times \text{LA} \rightarrow$<br>$=$<br>definition of $g \rightarrow g \triangleq [\neg 0, (1 \times \text{sz})^2 \text{ opb } \pi_1]$                       |               |
| $A + \text{Int} \times \text{LA} \times \text{Int} \times \text{LA} \rightarrow$<br>$\sqsubseteq$<br>definition of $\text{opb}$ , and $+$ monotonic                                                             |               |
| $A + \text{Int} \times \text{LA} \times \text{Int} \times \text{LA} \rightarrow$<br>$\sqsubseteq$<br>$\leq$ reflexive, so $\text{zero} \sqsubseteq \text{zero} \leq$ ; coproducts                               |               |
| $F(\text{Int} \times \text{LA}) \rightarrow \boxed{g} \rightarrow \boxed{\leq} \rightarrow \text{Int}$<br>$=$<br>definition of $g \rightarrow g \triangleq [\neg 0, (1 \times \text{sz})^2 \text{ opb } \pi_1]$ |               |

$$\text{mct}=(\text{single}\rightarrow\text{head tip},\langle\text{init col},\text{tail row}\rangle\text{mix}), \text{ Assoc}(\text{sb}) \Rightarrow (\mu X : \frac{[\text{wrap},\text{cat}]^\circ}{\mathfrak{g}} \text{P}([\text{tip},(X \times X) \text{bin}]) \text{est}(\text{R})) \sqsubseteq \frac{\text{flatten}^\circ}{\mathfrak{g}} \text{est}(\text{R}) \quad (9.3b)$$

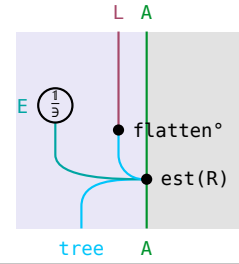
a least-cost bracketing of  $a_1 \oplus \dots \oplus a_n$  is read off an array holding one best tree per non-empty segment, each entry built from the column to its left and the row below it

## circuit

$$\text{LA} \rightarrow \frac{1}{\mathfrak{g}} \rightarrow \text{P}(\text{LA}) \rightarrow \text{E}(\text{flatten}^\circ) \rightarrow \text{P}(\text{tree}(\text{A})) \rightarrow \text{est}(\text{R}) \rightarrow \text{tree}(\text{A})$$

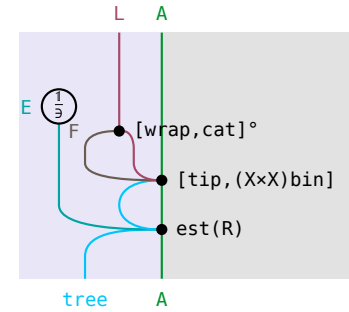
the specification —  $\text{flatten}(\text{tip}(\text{a})) = [\text{a}]$ ,  $\text{flatten}(\text{bin}(\text{l}, \text{r})) = \text{cat}(\text{flatten}(\text{l}), \text{flatten}(\text{r}))$

## Hinze–Marsden



$$\text{LA} \rightarrow \frac{1}{\mathfrak{g}} \rightarrow \text{P}(\text{LA}) \rightarrow \text{E}([\text{wrap},\text{cat}]^\circ) \rightarrow \text{PF}(\text{LA}) \rightarrow \text{P}([\text{tip},(X \times X)\text{bin}]) \rightarrow \text{P}(\text{tree}(\text{A})) \rightarrow \text{est}(\text{R}) \rightarrow \text{tree}(\text{A})$$

split the list in every way, bracket both halves, join. The condition is monotonicity **in context**,  $\sqsubseteq \text{F}(\text{Rn}(\text{flatten} \text{ flatten}^\circ)) \text{h} \sqsubseteq \text{hR}$  — only trees with the same flattening are compared — which is Proposition 9.3 at  $\text{H}^\circ = \text{flatten}$ , a map, with (9.5)  $[\text{tip},\text{bin}] \text{ cost} = (1 + (\text{cost}, \text{flatten})^2) \text{g}$  (the cost of a node reads only the cost and the flattening of its two subtrees) and (9.6)  $(1 + (\leq \times 1)^2) \text{g} \sqsubseteq \text{g} \leq (\text{g} \text{ monotonic on } \leq \text{ in its two cost arguments})$

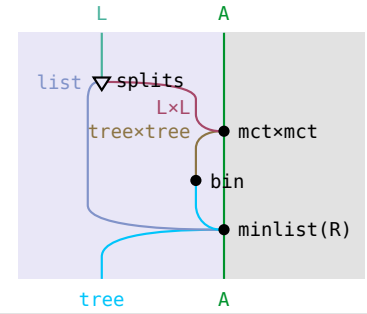


$$\text{LA} \rightarrow \frac{1}{\mathfrak{g}} \rightarrow \text{P}(\text{LA}) \rightarrow \text{E}(\text{cat}^\circ) \rightarrow \text{P}(\text{LA}^2) \rightarrow \text{P}((X \times X)\text{bin}) \rightarrow \text{P}(\text{tree}(\text{A})) \rightarrow \text{est}(\text{R}) \rightarrow \text{tree}(\text{A})$$

= Proposition 9.1: **wrap** and **cat** have disjoint ranges, and **single** is the coreflexive on singleton lists, where **wrap** returns

$$\text{LA} \rightarrow \text{splits} \rightarrow [\text{LA}^2] \rightarrow \text{list}((\text{mct} \times \text{mct})\text{bin}) \rightarrow [\text{tree}(\text{A})] \rightarrow \text{minlist}(\text{R}) \rightarrow \text{tree}(\text{A})$$

$\sqsubseteq \text{splits} \triangleq \langle \text{inits}^+, \text{tails}^+ \rangle$  zip implements  $\frac{\text{cat}^\circ}{\mathfrak{g}}$  and  $\text{minlist}(\text{R})$  implements  $\text{est}(\text{R})$ . Exponential, since the segments of one list overlap



$$\text{mct}=(\text{single}\rightarrow\text{head tip},\langle\text{init col},\text{tail row}\rangle\text{mix}) \quad (9.7)$$

= the tabulation:  $\text{mct } \text{xs}$  is needed for every non-empty segment  $\text{xs}$ , so the values are held as an array of rows,  $\text{array} \triangleq \text{inits list}(\text{row})$ ,  $\text{row} \triangleq \text{tails list}(\text{mct})$ ,  $\text{col} \triangleq \text{inits list}(\text{mct})$ ,  $\text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(\text{R})$

$$\text{col}=(\text{single}\rightarrow\text{head tip wrap},\langle\text{init col},\text{tail row}\rangle\text{next}) \quad (9.8)$$

$$\text{cons col}=(1 \times \text{array}) \text{ process} \quad (9.9)$$

$$\text{row}=(\text{single}\rightarrow\text{head tip wrap},\langle\text{mct},\text{tail row}\rangle\text{cons}) \quad (9.10)$$

a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop,  $\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}$ ,  $\text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})$

$$\text{array}=(\text{fstcol},\text{addcol}), \text{ fstcol} \triangleq \text{tip wrap wrap}$$

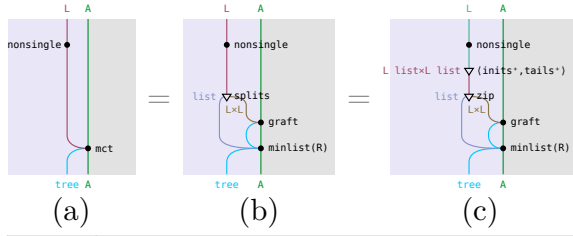
$$\text{addcol} \triangleq (\pi_1 \text{ tip wrap}, \text{step}) \text{ cons}, \text{ step} \triangleq (\text{process tail}, \pi_2) \text{ zip list}(\text{cons})$$

the program: one fold building the array column by column, cubic in the length of the input

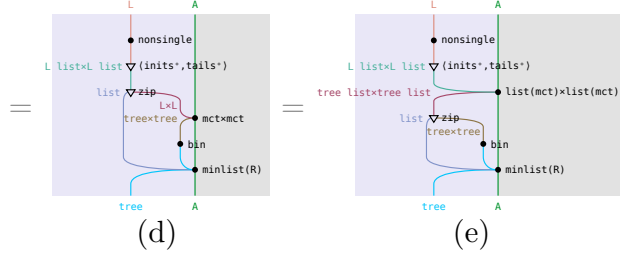
**nonsingle mct = nonsingle (init col,tail row) mix**

(9.7)

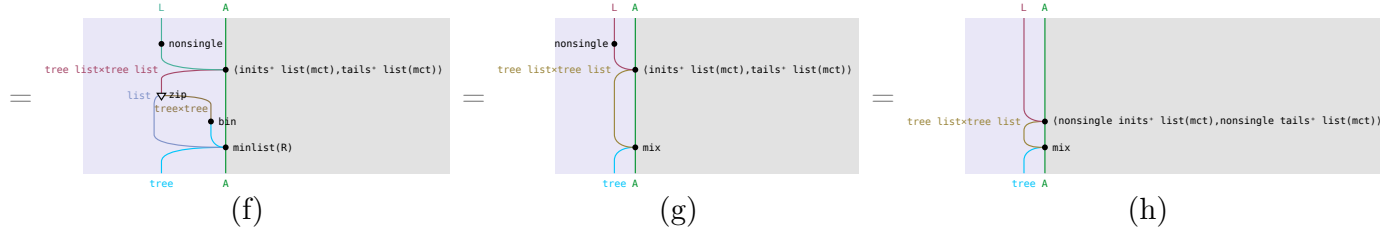
on a list of two or more elements, **mct** gives what **mix** gives on the column of its **init** and the row of its **tail**



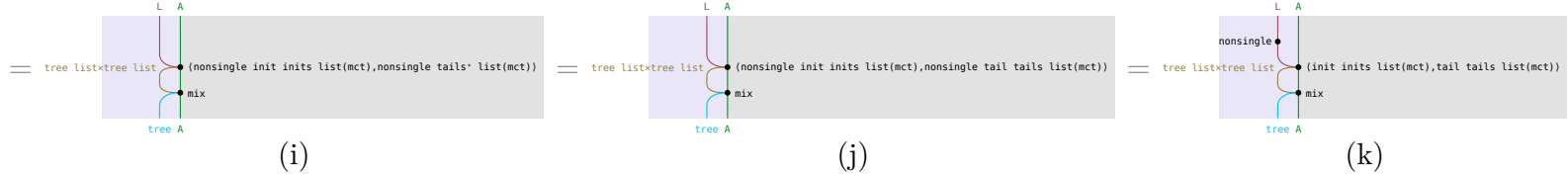
|     |                                                                                   |                                                         |
|-----|-----------------------------------------------------------------------------------|---------------------------------------------------------|
| (a) | nonsingle mct                                                                     | nonsingle mct = nonsingle splits list(graft) minlist(R) |
| (b) | = nonsingle splits list(graft) minlist(R)                                         | splits = (inits <sup>+</sup> , tails <sup>+</sup> ) zip |
| (c) | = nonsingle (inits <sup>+</sup> , tails <sup>+</sup> ) zip list(graft) minlist(R) | graft $\triangleq$ (mct $\times$ mct) bin               |



|     |                                                                                                                |                                                             |
|-----|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| (d) | = nonsingle (inits <sup>+</sup> , tails <sup>+</sup> ) zip list(mct $\times$ mct) list(bin) minlist(R)         | zip list(f $\times$ g) M = (list(f) $\times$ list(g)) zip M |
| (e) | = nonsingle (inits <sup>+</sup> , tails <sup>+</sup> ) (list(mct) $\times$ list(mct)) zip list(bin) minlist(R) | (R,S) (U $\times$ V) = (R U, S V)                           |



|     |                                                                                                   |                                                         |
|-----|---------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| (f) | = nonsingle (inits <sup>+</sup> list(mct), tails <sup>+</sup> list(mct)) zip list(bin) minlist(R) | mix = zip list(bin) minlist(R)                          |
| (g) | = nonsingle (inits <sup>+</sup> list(mct), tails <sup>+</sup> list(mct)) mix                      | nonsingle (U,V) = (nonsingle U, nonsingle V)            |
| (h) | = (nonsingle inits <sup>+</sup> list(mct), nonsingle tails <sup>+</sup> list(mct)) mix            | nonsingle inits <sup>+</sup> M = nonsingle init inits M |



|     |                                                                                |                                                         |
|-----|--------------------------------------------------------------------------------|---------------------------------------------------------|
| (i) | = (nonsingle init inits list(mct), nonsingle tails <sup>+</sup> list(mct)) mix | nonsingle tails <sup>+</sup> M = nonsingle tail tails M |
| (j) | = (nonsingle init inits list(mct), nonsingle tail tails list(mct)) mix         | nonsingle (U,V) = (nonsingle U, nonsingle V)            |
| (k) | = nonsingle (init inits list(mct), tail tails list(mct)) mix                   | col = inits list(mct)                                   |

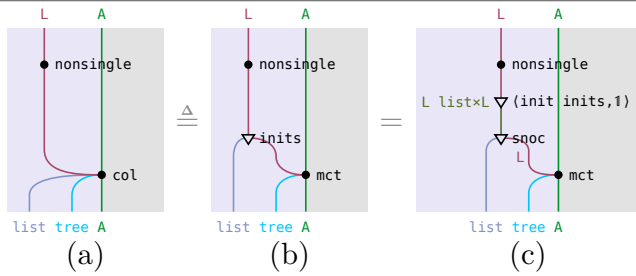


|     |                                                  |                       |
|-----|--------------------------------------------------|-----------------------|
| (l) | = nonsingle (init col, tail tails list(mct)) mix | row = tails list(mct) |
| (m) | = nonsingle (init col, tail row) mix             |                       |

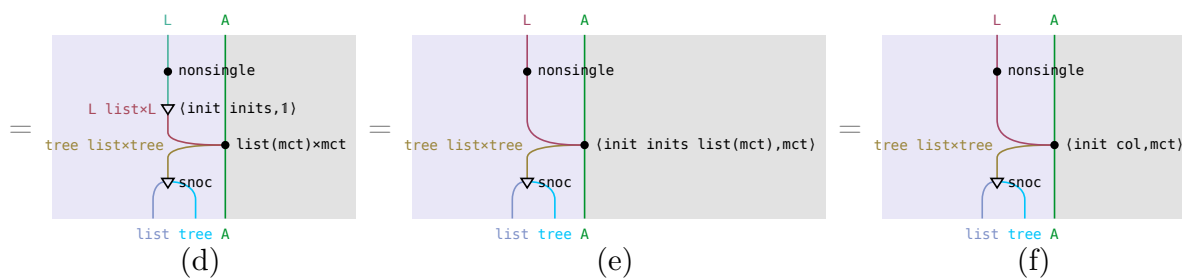
**nonsingle col = nonsingle (init col,tail row) next**

(9.8)

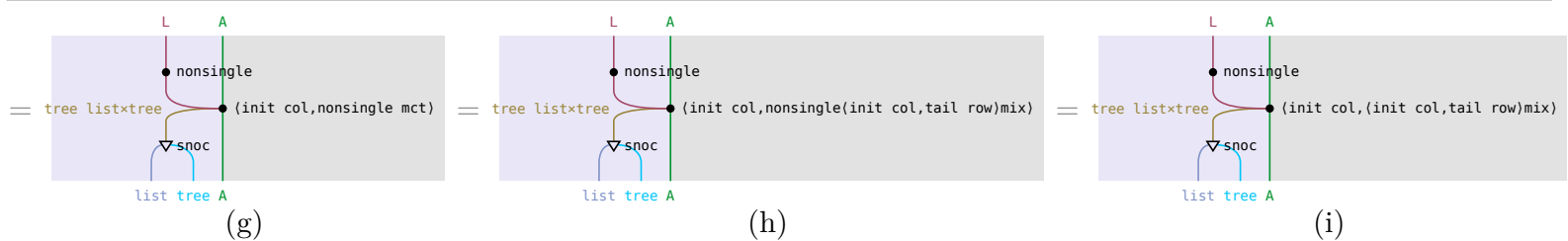
on a list of two or more elements, the column is what **next** makes of the column of its **init** and the row of its **tail**



|     |                                           |                                                        |
|-----|-------------------------------------------|--------------------------------------------------------|
| (a) | nonsingle col                             | col $\triangleq$ inits list(mct)                       |
| (b) | $\triangleq$ nonsingle inits list(mct)    | nonsingle inits M = nonsingle (init inits,1)<br>snoc M |
| (c) | = nonsingle (init inits,1) snoc list(mct) | snoc list(f) = (list(f) $\times$ f) snoc               |



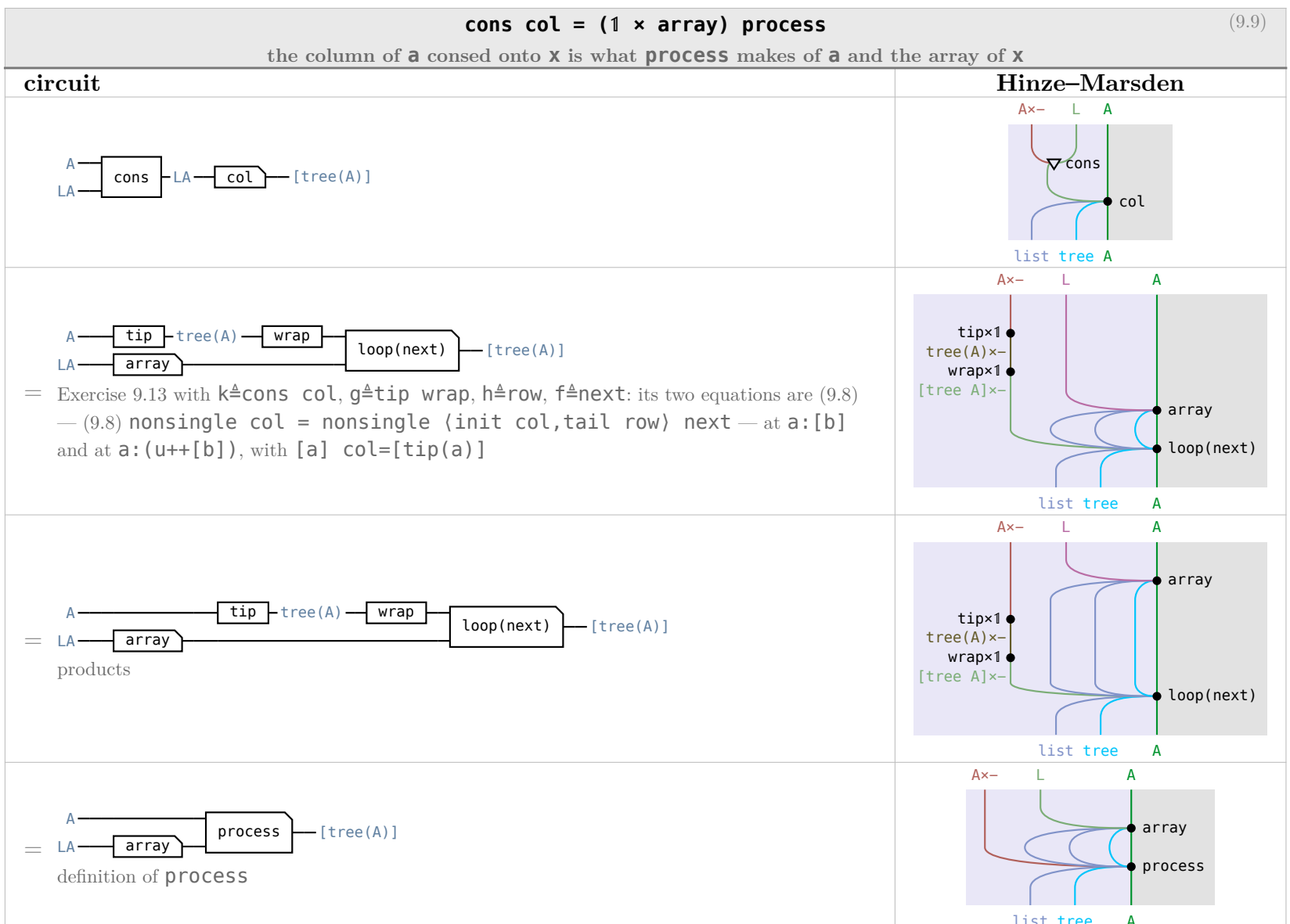
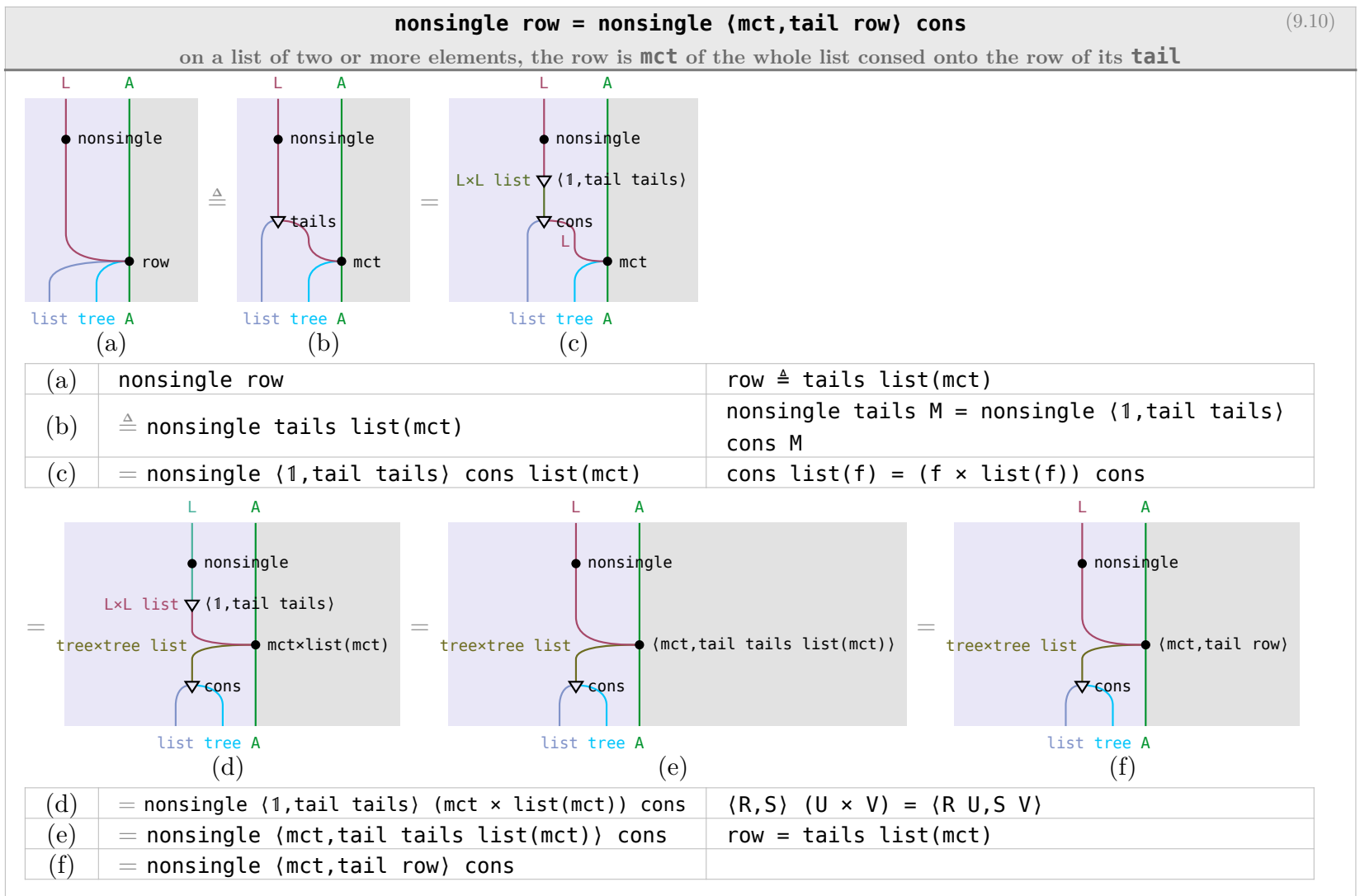
|     |                                                          |                                                                                |
|-----|----------------------------------------------------------|--------------------------------------------------------------------------------|
| (d) | = nonsingle (init inits,1) (list(mct) $\times$ mct) snoc | $\langle R,S \rangle (U \times V) = \langle R U, S V \rangle$                  |
| (e) | = nonsingle (init inits list(mct),mct) snoc              | col = inits list(mct)                                                          |
| (f) | = nonsingle (init col,mct) snoc                          | nonsingle $\langle U,V \rangle =$ nonsingle $\langle U,$ nonsingle $V \rangle$ |

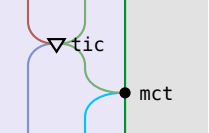
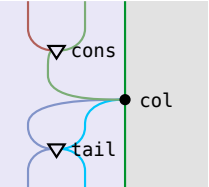
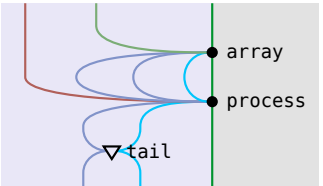
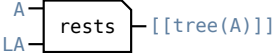
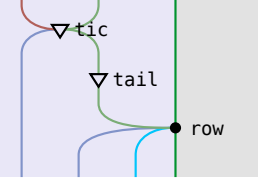


|     |                                                               |                                                                                                           |
|-----|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| (g) | = nonsingle (init col,nonsingle mct) snoc                     | (9.7) nonsingle mct = nonsingle (init col,tail row) mix                                                   |
| (h) | = nonsingle (init col,nonsingle (init col,tail row) mix) snoc | nonsingle $\langle U,V \rangle =$ nonsingle $\langle U,$ nonsingle $V \rangle$                            |
| (i) | = nonsingle (init col,(init col,tail row) mix) snoc           | $\langle \text{init col}, V \rangle (\pi_1, M) = \langle \text{init col}, (\text{init col}, V) M \rangle$ |



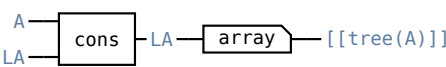
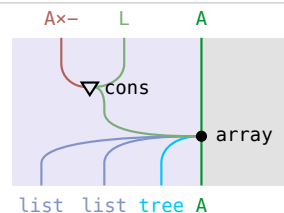
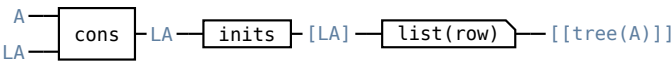
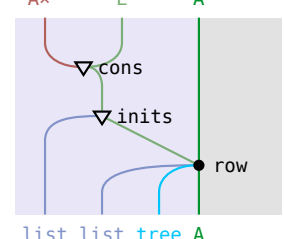
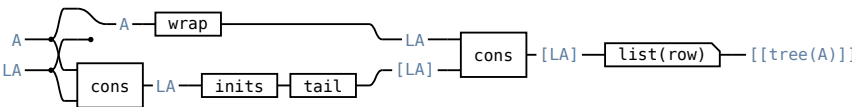
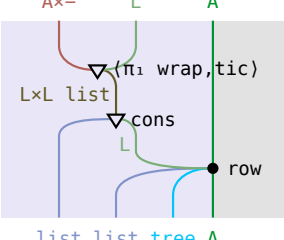
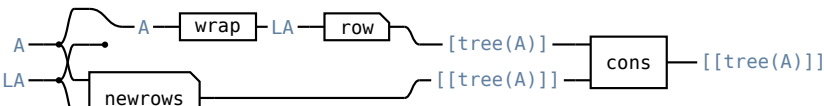
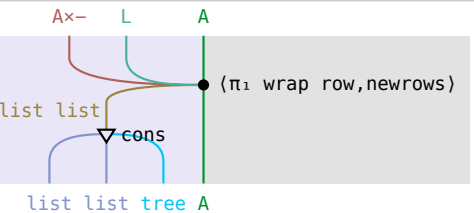
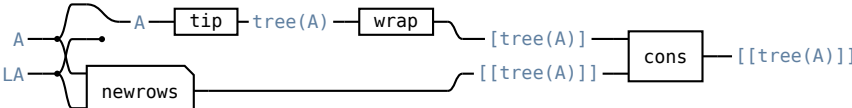
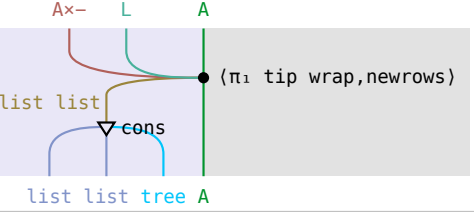
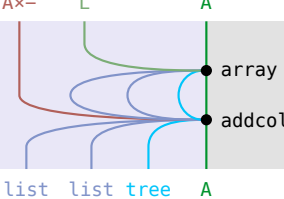
|     |                                                                          |                                                 |
|-----|--------------------------------------------------------------------------|-------------------------------------------------|
| (j) | = nonsingle (init col,tail row) $\langle \pi_1, \text{mix} \rangle$ snoc | next = $\langle \pi_1, \text{mix} \rangle$ snoc |
| (k) | = nonsingle (init col,tail row) next                                     |                                                 |



| <b>tops = (1 × array) process tail</b> <span style="float: right;">(9.3c)</span><br>the trees at the top of the new rows are what <b>process</b> makes of <b>a</b> and the array of <b>x</b> , less its first        |                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                              | Hinze–Marsden                                                                                             |
|                                                                                                                                     | <br>list tree A        |
| $= \begin{array}{c} A \\ LA \end{array} \text{cons} \text{LA} \text{inits} \text{tail} \text{LA} \text{list(mct)} \text{tree(A)}$ <p>definition of tops<math>\triangleq</math>tic list(mct), our name</p>            | <br>list tree A        |
| $= \begin{array}{c} A \\ LA \end{array} \text{cons} \text{LA} \text{col} \text{tail} \text{tree(A)}$ <p>tail list(f)=list(f) tail; definition of col</p>                                                             | <br>list tree A        |
| $= \begin{array}{c} A \\ LA \end{array} \text{array} \text{process} \text{tail} \text{tree(A)}$ <p>(9.9) — (9.9) cons col = (1 × array) process</p>                                                                  | <br>list tree A       |
| <b>rests = <math>\pi_2</math> array</b> <span style="float: right;">(9.3d)</span><br>the rest of each new row is the array of <b>x</b>                                                                               |                                                                                                           |
| circuit                                                                                                                                                                                                              | Hinze–Marsden                                                                                             |
|                                                                                                                                   | <br>list list tree A |
| $= \begin{array}{c} A \\ LA \end{array} \text{cons} \text{LA} \text{inits} \text{tail} \text{LA} \text{list(tail row)} \text{tree(A)}$ <p>definition of rests<math>\triangleq</math>tic list(tail row), our name</p> | <br>list list tree A |
| $= \begin{array}{c} A \\ LA \end{array} \text{array} \text{tree(A)}$ <p>tic list(tail)=<math>\pi_2</math> inits; definition of array</p>                                                                             | <br>list list tree A |



| $\text{newrows} = (1 \times \text{array}) \text{ step}$ <span style="float: right;">(9.3e)</span><br>the rows of the array of $\mathbf{a}:\mathbf{x}$ below its first are what <b>step</b> makes of $\mathbf{a}$ and the array of $\mathbf{x}$                                                                                                           |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                  | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                                                                          |               |
| $= \text{LA} \rightarrow \text{cons} \rightarrow \text{LA} \rightarrow \text{inits} \rightarrow \text{tail} \rightarrow [\text{LA}] \rightarrow \text{list}(\text{row}) \rightarrow [[\text{tree}(\text{A})]]$ <p>definition of <math>\text{newrows} \triangleq \text{tic list}(\text{row})</math>, our name</p>                                         |               |
| $= \text{LA} \rightarrow \text{cons} \rightarrow \text{LA} \rightarrow \text{inits} \rightarrow \text{tail} \rightarrow [\text{LA}] \rightarrow \text{list}(\text{mct, tail row cons}) \rightarrow [[\text{tree}(\text{A})]]$ <p>(9.10) — (9.10) nonsingle row = nonsingle (mct, tail row)<br/> cons — on the non-singleton tic lists</p>                |               |
| $= \text{LA} \rightarrow \text{tops} \rightarrow \text{rests} \rightarrow \text{zip} \rightarrow [\text{tree}(\text{A}) \times [\text{tree}(\text{A})]] \rightarrow \text{list}(\text{cons}) \rightarrow [[\text{tree}(\text{A})]]$ <p><math>\text{list}(f,g) = (\text{list}(f), \text{list}(g))</math> zip; products; definitions of tops and rests</p> |               |
| $= \text{LA} \rightarrow \text{array} \rightarrow \text{step} \rightarrow [[\text{tree}(\text{A})]]$ <p><math>\text{tops} = (1 \times \text{array})</math> process tail and rests = <math>\pi_2</math> array;<br/> products; definition of step</p>                                                                                                      |               |

| $\text{cons array} = (1 \times \text{array}) \text{ addcol}$ <span style="float: right;">(9.3f)</span><br>the array of <b>a</b> consed onto <b>x</b> is what <b>addcol</b> makes of <b>a</b> and the array of <b>x</b> |                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                | Hinze–Marsden                                                                         |
|                                                                                                                                       |    |
| $=$ <br>definition of <b>array</b>                                                                                                    |    |
| $=$ <br>cons inits= $\langle \pi_1 \text{ wrap, tic} \rangle$ cons, abbreviating cons inits tail by tic                              |    |
| $=$ <br>cons list(f)=( $f \times \text{list}(f)$ ) cons; definition of <b>newrows</b>                                              |   |
| $=$ <br>wrap row=tip wrap                                                                                                          |  |
| $=$ <br>newrows = ( $1 \times \text{array}$ ) step ; products; definition of <b>addcol</b>                                          |  |

## §9.4. Data compression

(9.4a)

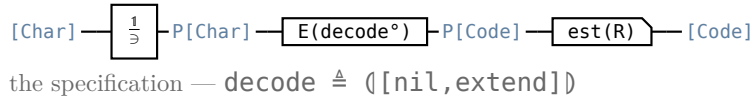
|    | name   | type                                                           | definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | meaning                                                                                                          |
|----|--------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1  | String | Type                                                           | $\text{String} \triangleq [\text{Char}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | a string is a snoc-list of characters                                                                            |
| 2  | LA     | Type                                                           | $\text{LA} \triangleq [\text{A}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | a non-empty snoc-list: one value, then any number more appended                                                  |
| 3  | Code   | Type                                                           | $\text{Code} \triangleq \text{sym}(\text{Char}) \mid \text{ptr}(\text{String}, \text{String})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | a code is one character, or a pointer back to a repeated piece                                                   |
| 4  | extend | $\text{String} \times \text{Code} \rightarrow [\text{Char}]$   | $\text{extend} \triangleq \text{extend} \ , \ \text{extend}((\text{xs}, \text{sym}(\text{a})), \text{x}) = \text{x} = \text{snoc}(\text{xs}, \text{a}) \ , \ \text{extend}((\text{xs}, \text{ptr}(\text{ys}, \text{zs})), \text{x}) = \text{x} = \text{cat}(\text{xs}, \text{zs}) \wedge \text{zs} \neq [] \wedge (\text{init}^+)(\text{cat}(\text{ys}, \text{zs}), \text{cat}(\text{xs}, \text{zs}))$                                                                                                                                                                                                                                                                                                                  | a symbol appends its character; a pointer appends <b>zs</b> when <b>ys#zs</b> is a proper prefix of <b>xs#zs</b> |
| 5  | decode | $[\text{Code}] \rightarrow [\text{Char}]$                      | $\text{decode} \triangleq ([\text{nil}, \text{extend}])$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | rebuild the string from its codes, left to right; <b>H=decode°</b>                                               |
| 6  | bytes  | $\text{Code} \rightarrow \mathbb{N}$                           | $\text{bytes}(\text{sym}(\text{a})) = \text{c} \ , \ \text{bytes}(\text{ptr}(\text{a}, \text{a}')) = \text{p}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | a symbol costs <b>c</b> , a pointer costs <b>p</b>                                                               |
| 7  | size   | $[\text{Code}] \rightarrow \mathbb{N}$                         | $([\text{nil}, \text{distr} [1 \times \text{c}, 1 \times \text{p}] \text{ plus}]) = \text{size} \ , \ \text{size}(\text{wrap}(\text{a})) = 0 \ , \ \text{size}(\text{snoc}(\text{cs}, \text{e})) = \text{size}(\text{cs}) + \text{bytes}(\text{e})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | the total cost of a code sequence                                                                                |
| 8  | R      | $[\text{Code}] \rightarrow [\text{Code}]$                      | $\text{R} = \text{size} \leq \text{size}^\circ \ , \ \text{R}(\text{u}, \text{v}) \iff \text{size}(\text{u}) \leq \text{size}(\text{v})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>u</b> costs no more than <b>v</b>                                                                             |
| 9  | prefix | $[\text{Char}] \rightarrow [\text{Char}]$                      | $\text{x prefix ys} \iff (\text{prefix}^\circ)(\text{ys}, \text{x})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>ys</b> is a prefix of <b>x</b>                                                                                |
| 10 | (T+T)  | $\text{Code} \rightarrow \text{Code}$                          | $(\text{T}+\text{T}) = [\text{sym}, \text{ptr}]^\circ \ (\text{T} + \text{T}) \ [\text{sym}, \text{ptr}] \ , \ (\text{T}+\text{T}) (\text{sym}(\text{a}), \text{sym}(\text{a}')) = \text{true} \ , \ (\text{T}+\text{T}) (\text{ptr}(\text{a}, \text{a}'), \text{ptr}(\text{a}'', \text{a}''')) = \text{true} \ , \ (\forall \text{a}. \forall \text{a}'. \text{e} = \text{sym}(\text{a}) \implies \text{f} = \text{sym}(\text{a}') \implies \text{false}) \wedge (\forall \text{a}. \forall \text{a}'. \forall \text{a}'''. \text{e} = \text{ptr}(\text{a}, \text{a}') \implies \text{f} = \text{ptr}(\text{a}'', \text{a}''') \implies \text{false}) \implies (\text{T}+\text{T})(\text{e}, \text{f}) = \text{false}$ | any symbol to any symbol, any pointer to any pointer                                                             |
| 11 | Q      | $\text{F}[\text{Char}] \rightarrow \text{F}[\text{Char}]$      | $\text{Q} = \text{F}((\text{T}+\text{T}), \text{prefix}^\circ) \ , \ \text{Q}(\text{l}(\text{val}), \text{l}(\text{val}')) = \text{true} \ , \ \text{Q}(\text{r}(\text{q}), \text{r}(\text{r})) = (\text{prefix}^\circ)(\pi_1(\text{q}), \pi_1(\text{r})) \wedge (\text{T}+\text{T})(\pi_2(\text{q}), \pi_2(\text{r})) \ , \ (\forall \text{val}. \forall \text{val}'. \text{u} = \text{l}(\text{val}) \implies \text{v} = \text{l}(\text{val}') \implies \text{false}) \wedge (\forall \text{q}. \forall \text{r}. \text{u} = \text{r}(\text{q}) \implies \text{v} = \text{r}(\text{r}) \implies \text{false}) \implies \text{Q}(\text{u}, \text{v}) = \text{false}$                                                   | compare two decompositions by the prefix order on the strings and <b>U</b> on the codes                          |
| 12 | reduce | $[\text{Char}] \rightarrow [\text{String} \times \text{Code}]$ | $\text{reduce} \triangleq \text{reduce} \ , \ \text{q} \in \text{reduce}(\text{w}) \iff \text{extend}(\text{q}, \text{w})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | the last code of a string: one symbol, or a pointer to its longest repeated tail                                 |

$$(\mu X : \frac{[nil, extend]^{\circ}}{\mathfrak{S}} \text{thin}(Q) P([nil, (X \times 1) \text{snoc}]) \text{est}(R)) \sqsubseteq \frac{\text{decode}^{\circ}}{\mathfrak{S}} \text{est}(R), \quad (9.4b)$$

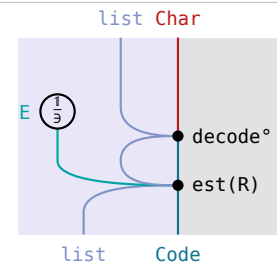
$$\text{encode} = (\text{null} \rightarrow \text{nil}, \text{reduce list}((\text{encode} \times 1) \text{snoc}) \text{minlist}(R))$$

a smallest code sequence decoding to the given string is built from the right, each step emitting the last character as a symbol or ending with a pointer back into what has already been decoded

### circuit

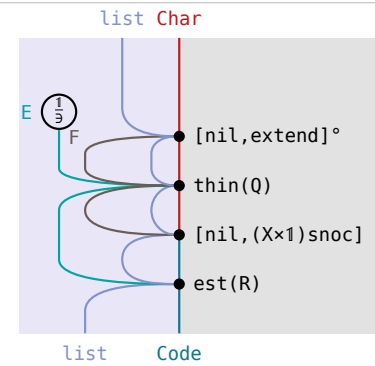


### Hinze–Marsden

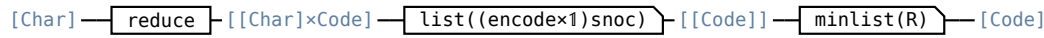
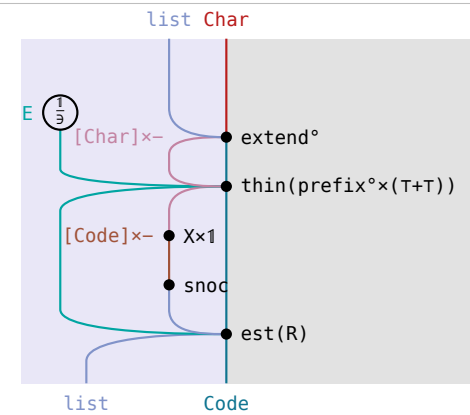


at  $Q \triangleq F(T+T, \text{prefix}^{\circ}) = 1 + (\text{prefix}^{\circ} \times (T+T))$ , the two  $T$  on symbols and on pointers. Monotonicity  $F(R) \alpha \sqsubseteq \alpha R$  is routine, the two costs being constants:  $(T+T)[c, p] = [c, p]$ .

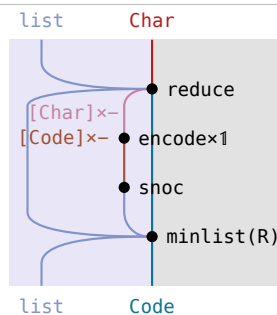
Proposition 9.4 splits the thinning condition in two —  $F(T+T, R) \alpha \sqsubseteq \alpha R$ , and  $\text{decode prefix} \sqsubseteq R^{\circ} \text{decode}$ , for which  $\text{decode init} \sqsubseteq R^{\circ} \text{decode}$  is enough: dropping the last character of the output shortens or removes the last code element and never raises the cost. So between a symbol and a pointer nothing can be decided in advance, and between two pointers the longer match wins



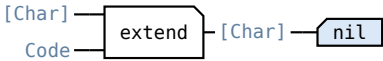
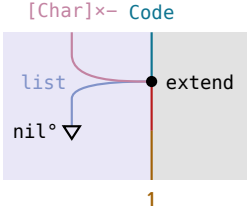
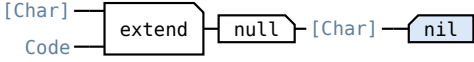
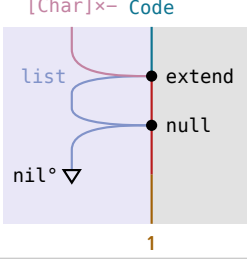
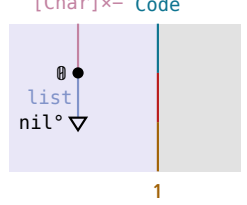
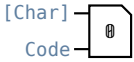
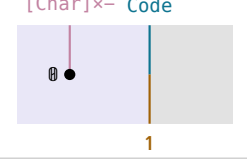
Proposition 9.1:  $\text{nil}$  and  $\text{extend}$  have disjoint ranges. The decompositions of one string are  $\frac{\text{extend}^{\circ}}{\mathfrak{S}} (ws \# [a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr } (ys, zs)) \mid xs \# zs = ws \# [a], ys \# zs \text{ a proper prefix of } ws\}$  — take the last character as a symbol, or end with a pointer



$\text{reduce}$  implements  $\frac{\text{extend}^{\circ}}{\mathfrak{S}} \text{thin}(\text{prefix}^{\circ} \times (T+T))$ : thinning leaves at most two, the symbol and the pointer of the longest repeated tail  $\text{lrt}$ . Again exponential; the book gives no tabulation for it



| $Q \vdash F(\text{decode}^\circ) [\text{nil}, \text{snoc}] \sqsubseteq F(\text{decode}^\circ) [\text{nil}, \text{snoc}] R$ (9.4c)                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| every code sequence built from a decomposition $Q$ puts above a given one — a code element of the same kind, a longer front string — costs at least as much as one built from the given one                                                                                                                                                                                                                                                                                                                                                                            |               |
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
| $=$ <p> <math>Q \triangleq 1 + (\text{prefix}^\circ \times (T+T))</math> is <math>F(T+T, \text{prefix}^\circ) - Q(l(\text{val}), l(\text{val}')) = \text{true}</math>,<br/> <math>Q(r(q), r(r)) = (\text{prefix}^\circ)(\pi_1(q), \pi_1(r)) \wedge (T+T)(\pi_2(q), \pi_2(r))</math>, <math>(\forall \text{val}. \forall \text{val}'. u = l(\text{val}) \Rightarrow v = l(\text{val}') \Rightarrow \text{false}) \wedge (\forall q. \forall r. u = r(q) \Rightarrow v = r(r) \Rightarrow \text{false})</math><br/> <math>\Rightarrow Q(u, v) = \text{false}</math> </p> |               |
| $=$ <p>the bifunctor <math>F</math> preserves composition: <math>F(U, \text{prefix}^\circ) F(1, \text{decode}^\circ) = F(U, \text{prefix}^\circ \text{decode}^\circ)</math></p>                                                                                                                                                                                                                                                                                                                                                                                        |               |
| $\sqsubseteq$ <p>Proposition 9.4's second condition <math>\text{prefix}^\circ \text{decode}^\circ \sqsubseteq \text{decode}^\circ R</math>: dropping the last character drops the last <b>sym</b>, or shortens or drops the last pointer. Proved at <b>prefix</b> directly by induction, not through <b>init</b></p>                                                                                                                                                                                                                                                   |               |
| $=$ <p><math>F</math> preserves composition: <math>F(U, \text{decode}^\circ R) = F(1, \text{decode}^\circ) F(U, R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| $\sqsubseteq$ <p>Proposition 9.4's first condition <math>F(T+T, R) \alpha \sqsubseteq \alpha R</math>, left as an exercise in the book: <b>snoc</b> adds the cost of the last element to both sides, and <math>[c, p](T+T) = [c, p]</math></p>                                                                                                                                                                                                                                                                                                                         |               |

| <div> <div>extend nil<sup>°</sup> = 0</div> <div>(9.4d)</div> </div> <div>no string is both a result of <b>extend</b> and the result [] of <b>nil</b></div>                                                      |                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                          | Hinze–Marsden                                                                        |
|                                                                                                                                 |   |
|  <p><math>\text{nil}</math> returns only [], so <math>\text{nil}^\circ = \text{null } \text{nil}^\circ</math></p>               |   |
|  <p><math>\text{extend } \text{null} = 0</math>: a symbol ends the string with a character, a pointer with its non-empty ZS</p> |   |
|  <p>0 composed with anything is 0</p>                                                                                          |  |